

CHAPTER 1

DIGITAL ELECTRONICS

Syllabus: Digital electronics: logic functions, minimization, design and synthesis of combinational and sequential circuits, number representation and computer arithmetic (fixed and floating point).

1.1 INTRODUCTION

Digital electronics represents discrete signals instead of signals in a continuous range. It uses two binary levels 0's (corresponding to false) and 1's (corresponding to true). The main reason for advancement in digital electronics is integrated circuits (ICs).

1.2 NUMBER SYSTEM

Number system is an age old method to represent numerals (Fig. 1.1). Decimal number system is the most common number system and binary number system is used by computers.

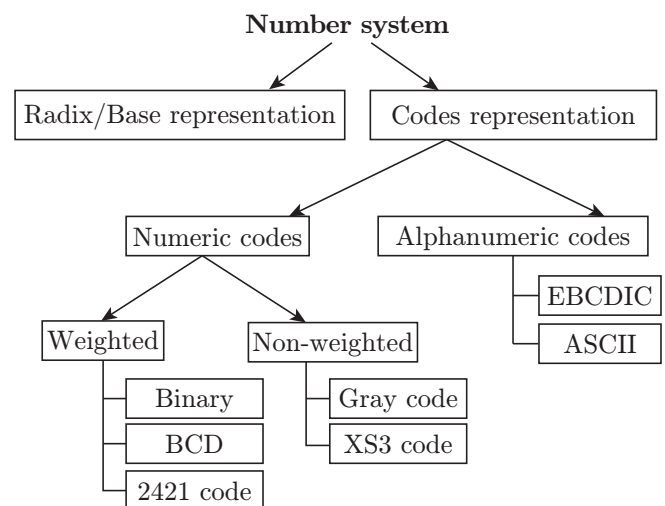


Figure 1.1 | Number system.

Table 1.1 | Types of number systems

Base/Radix	Unique Numbers	Terminology	Example
10	0 to 9	Decimal number system	10.47
8	0 to 7	Octal number system	65.32
2	0 to 1	Binary number system	110.11
16	0 to 15 0 to 9 are numerals and 10 to 15 are represented with alphabets 10 = A, 11 = B, 12 = C, ... 15 = F	Hexadecimal number system	BAD.1A

Radix or base is the number of unique digits, so the different base or radix can be expressed as represented in Table 1.1. For example, $(57)_{10}$ – here 10 is the base (or radix) and 57 is the number.

Example 1.1

Let us understand the logic of representation by considering $(57.1)_{10}$. We know 57 means $50 + 7 + (1/10)$. Now, representing the same by making use of radix (which is 10), we get

$$5 \times (10)^1 + 7 \times (10)^0 + 1 \times (10)^{-1}$$

1.2.1 Conversions of Number System

A number of a particular number system can be converted into another number system, as follows:

- 1. Decimal number system to any number system:** Consider the example of $(57.3)_{10}$. To find the binary equivalent, we have

2 57			
2 28	1		
2 14	0	$2 \times 0.3 = 0.6$	0
2 7	0	$2 \times 0.6 = 1.2$	1
2 3	1	$2 \times 0.2 = 0.4$	0
2 1	1	$2 \times 0.4 = 0.8$	0
(111001) ₂		(0.0100) ₂	

Therefore, $(57.3)_{10} = (111001.0100)_2$

- 2. Any number system to decimal number system:** Consider the example of $(111001.0100)_2$. To find the decimal equivalent we have

$$(111001.0100)_2 = 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1}$$

$$+ 1 \times 2^{-2} + 0 \times 2^{-3} + 0 \times 2^{-4} \\ = 32 + 16 + 8 + 0 + 0 + 1 + 0 \\ + 0.25 + 0 + 0 = (57.25)_{10}$$

Another example is of $(101.11)_2 = (5.75)_{10} = (5.6)_8 = (5.C)_{16}$.

1.2.2 Complement of a Number

Any given number will have two complements:

- 1. Radix minus one complement (R-1's complement):** It is obtained by subtracting the given number from the highest possible number. For example, if there is a two-digit decimal number, it will be subtracted from 99; whereas a 3-bit number in a binary system will be subtracted from 111.

Note: The 1's complement of a binary number is obtained by interchanging the 1's and 0's. For example, 1's complement of $(1011)_2$ is $(0100)_2$.

- 2. Radix complement (R's complement):** This is also known as true complement. It is obtained by adding one to R-1's complement. For example, $(57)_{10}$ – its 9's complement is 42 and its 10's complement is 43.

1.2.2.1 Subtraction Using R-1's Complement

For the following subtractions,

$$\begin{array}{r} A \\ -B \\ \hline \hline \end{array}$$

the steps involved are as follows:

Step 1. Find R-1's complement of B.

Step 2. Add A and B.

Step 3. If there is carry, then add this carry to the answer.

If there is no carry, then find the R-1's complement of the answer and place a negative sign in front of the answer.

Problem 1.1: $(101)_2 - (011)_2$

Solution:

Step 1. 1's Complement of 011 is 100.

Step 2. The addition of these two numbers is $(101 + 100 = 1\ 001)$.

Step 3. There is a carry of 1, so now we will add this carry to the answer $(1 + 001)$, which is $(010)_2$.

Problem 1.2: $(011)_2 - (101)_2$

Solution:

Step 1. 1's Complement of 101 is 010.

Step 2. The addition of these two numbers is $(011 + 010 = 0\ 101)$.

Step 3. There is a no carry, so now we will find 1's complement of the answer obtained (which is 010) and place a negative sign in front of it. So, the answer is $(-010)_2$.

1.2.2.2 Subtraction Using R's Complement

In this method, we find R's complement of the subtrahend and we do not add the carry, as in step 2. Rest of the steps are the same.

Problem 1.3: $(101)_2 - (011)_2$

Solution:

Step 1. 2's complement of 011 is 101.

Step 2. The addition of these two numbers is $(101 + 101 = 1\ 010)$.

Step 3. There is a carry of 1, so now we will not add this carry to the answer (001), which is $(010)_2$.

Problem 1.4: $(011)_2 - (101)_2$

Solution:

Step 1. 2's complement of 101 is 011.

Step 2. The addition of these two numbers is $(011 + 011 = 0\ 110)$.

Step 3. There is a no carry, so now we will find 2's complement of the answer obtained (which is 010) and place a negative sign in front of it. So, the answer is $(-010)_2$.

1.2.3 Representation of Negative Numbers

In $-(57)_{10} = (1111001)$, the most significant bit is 1, showing that the number is negative binary number.

1.2.4 The IEEE Standard for Floating Point Numbers

The IEEE (Institute of Electrical and Electronics Engineers) has provided a standard for floating point numbers and their arithmetic. This standard specifies how single precision (32-bit) and double precision (64-bit) floating point numbers are to be represented.

1. Single Precision: The IEEE single precision floating point standard representation requires a 32-bit word. The first bit to the left is the sign bit, the next eight bits are the exponent bits, and the final 23 bits form the fraction part (Fig. 1.2).

The value of a normalized number $= (-1)^s \times 1.m \times 2^{e-127}$

where $s = 1$ for negative number, $s = 0$ for positive number, m = mantissa and e = exponent.

2. Double Precision: The IEEE double precision floating point standard representation requires a 64-bit word. The first bit is the sign bit, the next eleven bits are the exponent bits, and the final 52 bits form the fraction part (Fig. 1.3).

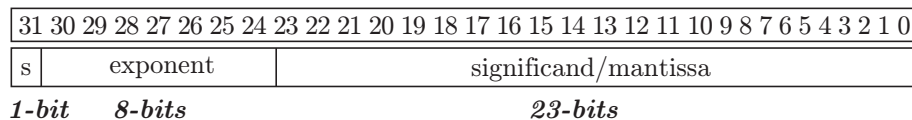


Figure 1.2 | Single Precision (32 bits).

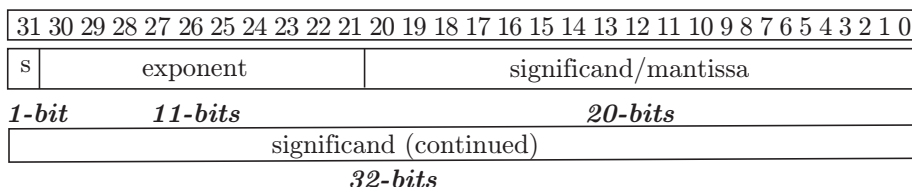


Figure 1.3 | Double Precision (64 bits).

Problem 1.5: Consider the decimal number: +116.25. Represent it using single precision IEEE floating point standard.

Solution: The equivalent binary representation is +1110100.01

Normalizing number = $+1.11010001 \times 2^6$
= $+1.11010001 \times 2^{133-127}$
= $+1.11010001$
 $\times 2^{10000101-01111111}$

In IEEE 754 format:

Sign	Exponent	Fraction
0	10000101	110100010000000000000000

1.3 BOOLEAN LOGIC

The concept of Boolean logic was proposed by George Boole, in which any information or data is represented in the form of two states, which are complement to each other (such as 0 and 1, hot and cold, black and white, north and south, etc.).

Logic gates form the basic building block of the digital system. A logic gate is an electronic device that takes logical decisions based on given input combinations and

its output can be represented by Boolean expression. The symbols used to represent logic gates are shown in Fig. 1.4.

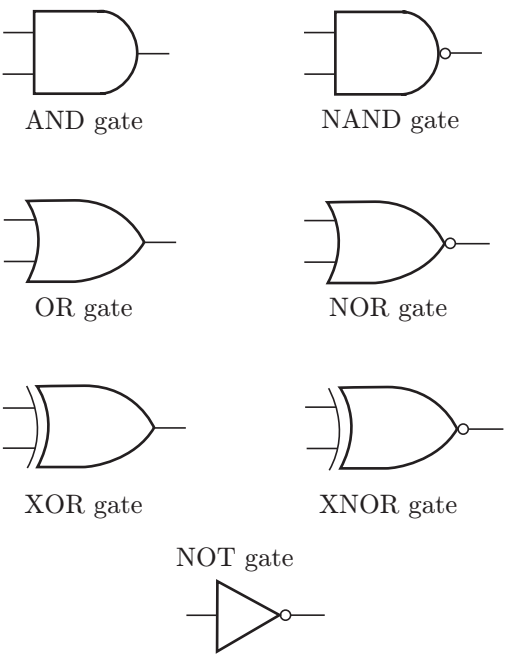


Figure 1.4 | Logic gates symbols.

Table 1.2 represents 15 basic operations that can be performed.

Table 1.2 | Basic operations in Boolean algebra

Inputs		Operations					
X	Y	Null	Identity	Inhibit		Transfer	
				X Inhibit Y	Y Inhibit X		
0	0	0	1	0	0	0	
0	1	0	1	0	1	1	
1	0	0	1	1	0	0	
1	1	0	1	0	0	1	

Inputs		Operations					
X	Y	AND	OR	NOT		XOR	XNOR
				NOT X	NOT Y		
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	0	1	1	0
1	1	1	1	0	0	0	1

(Continued)

Table 1.2 | Continued

Inputs		Operations			
X	Y	NAND	NOR	Implication	
				X Implication Y	Y Implication X
0	0	1	1	1	1
0	1	1	0	1	0
1	0	1	0	0	1
1	1	0	0	1	1

Table 1.2 is interpreted as follows:

1. Null – All the outputs are zero.
2. Identity – All the outputs are one.
3. X inhibit Y – Represented as $(x \cdot \sim y)$
4. Y inhibit X – Represented as $(y \cdot \sim x)$
5. Transfer – Same as y
6. AND – $x \cdot y$ (Similar to intersection in set theory)
7. OR – $x + y$ (Similar to union in set theory)
8. NOT X – $\sim x$ (Also known as complement of x)
9. NOT Y – $\sim y$ (Also known as complement of y)
10. XOR – $(\sim x \cdot y) + (x \cdot \sim y)$ (Also known as parity checker)
11. XNOR – $(x \cdot y) + (\sim x \cdot \sim y)$
12. NAND – $\sim(x \cdot y)$
13. NOR – $\sim(x + y)$
14. X implication Y ($\sim x + y$)
15. Y implication X ($\sim y + x$)

Problem 1.6: It is defined that $x * y = x'y' + xy$ and $z = x * y$. Among the following P , Q and R , which statement is correct?

$$P: x = y * z$$

$$Q: y = x * z$$

$$R: x * y * z = 1$$

Solution:

x	Y	z	P	Q	R
0	0	1	0	0	1
0	1	0	0	1	1
1	0	0	1	0	1
1	1	1	1	1	1

1.3.1 Boolean Algebra Laws (Huntington's Postulates)

The laws of Boolean Algebra are given by Huntington's postulates:

1. $X + 0 = X$
2. $X + (\sim X) = 1$

$$3. X + X = X$$

$$4. X + 1 = 1$$

$$5. X + Y = Y + X$$

$$6. X + (Y + Z) = (X + Y) + Z$$

$$7. X \cdot (Y + Z) = X \cdot Y + X \cdot Z$$

$$8. \sim(X + Y) = \sim X \cdot \sim Y \text{ (Also called DeMorgan's law)}$$

$$9. X + (X \cdot Y) = X$$

$$10. \sim(\sim X) = X$$

1.3.2 Duality Theorem

Every expression remains valid if we interchange the operator and identity elements. Following laws are the dual of the respective laws written above.

$$1. X \cdot 1 = X$$

$$2. (X \cdot \sim X) = X \cdot X' = 0$$

$$3. X \cdot X = X$$

$$4. X \cdot 0 = 0$$

$$5. X \cdot Y = Y \cdot X$$

$$6. X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$$

$$7. X + (Y \cdot Z) = (X + Y) \cdot (X + Z)$$

$$8. (X \cdot Y)' = X' + Y'$$

$$9. X \cdot (X + Y) = X$$

1.3.3 Consensus Theorem

It is applicable only if a Boolean function has

1. three variables
2. each variable is used two times
3. only one variable in complemented or non-complemented form

Then, the term related to the complemented and non-complemented variable is the answer.

Example 1.2

$$AB + A'C + BC = AB + A'C$$

1.3.4 Positive Logic and Negative Logic

Usually, the two (binary) levels are represented in two different ways:

1 (signifying high) and 0 (signifying low) – Known as positive logic

or

1 (signifying low) and 0 (signifying high) – Known as negative logic

1.3.4.1 Canonical and Standard Forms

The minterms and the maxterms are given in Table 1.3.

Product-of-sums (POS) form:

$$\begin{aligned} F &= (x'y'z') + (xy'z') + (xyz) \\ &= m_0 + m_4 + m_7 \\ &= \sum m(0, 4, 7) \end{aligned}$$

Sum-of-products (SOP) form:

$$\begin{aligned} F &= (X + Y + Z) \cdot (X' + Y + Z) \cdot (X + Y + Z) \\ &= M_0 \cdot M_4 \cdot M_7 \\ &= \prod M(0, 4, 7) \end{aligned}$$

In order to convert minterm to maxterm, put all those numbers in maxterm which are not present in minterm, and vice versa.

$$\sum m(0, 1, 5) = \prod M(2, 3, 4, 6, 7)$$

1.3.4.2 Minimization of Expression

A Boolean expression which cannot be further reduced is said to be in minimal form. Each term in minimal Boolean expression is called prime implicant. The following are the two methods for minimization:

- 1. Boolean Theorems (postulates):** Using the Boolean postulates, we can simplify an expression using algebraic method. This method is relatively difficult as we need to remember all the postulates.

Problem 1.7: Simplify the Boolean function $X = (A + B)(A + C)$.

Solution:

$$\begin{aligned} X &= AA + AC + AB + BC \\ &= A + AC + AB + BC \\ &= A(1 + C) + AB + BC \\ &= A(1) + AB + BC \\ &= A + AB + BC \\ &= A(1 + B) + BC \\ &= A(1) + BC \\ &= A + BC \end{aligned}$$

Problem 1.8: Simplify the Boolean function $X = A + A'B$.

Solution:

$$\begin{aligned} X &= A(1 + B) + A'B \\ &= A \cdot B + A + A'B = A \cdot B + A'B + A \\ &= (A + A') \cdot B + A = 1 \cdot B + A = B + A \end{aligned}$$

Table 1.3 | Minterms and Maxterms

Decimal	Binary	Minterms (SOP)		Maxterms (POS)	
		Term	Designation	Term	Designation
0	000	$x'y'z'$	m_0	$X + Y + Z$	M_0
1	001	$x'y'z$	m_1	$X + Y + Z'$	M_1
2	010	$x'y z'$	m_2	$X + Y' + Z$	M_2
3	011	$x'y z$	m_3	$X + Y' + Z'$	M_3
4	100	$xy'z'$	m_4	$X' + Y + Z$	M_4
5	101	$xy'z$	m_5	$X' + Y + Z'$	M_5
6	110	xyz'	m_6	$X' + Y' + Z$	M_6
7	111	xyz	m_7	$X' + Y' + Z'$	M_7

2. Karnaugh-Map Method (K-Map method):

Karnaugh-map (K-map) involves graphical representation for simplifying Boolean expressions. It consists of group of adjacent cells in which each cell corresponds to one of the combinations of max-terms or min-terms of n variables and is represented by group of literals. For n -variable K-map, there are 2^n cells required (Figs. 1.5–1.7). From one cell to the neighbouring cell there must be only one literal change. While solving the K-map from higher-order group to lower-order group only, the simplification must be done. In K-map, leftmost column and rightmost column of the same row are considered to be adjacent.

If $K = 1$, it represents that all the cells are 1 so there is no need of circuit.

1.3.4.3 Grouping of Cells for Simplification

The grouping of cells for simplification is done in the following ways:

1. **Pairs:** If two adjacent cells either in row or in column contain 1's, they are said to be in pairs.

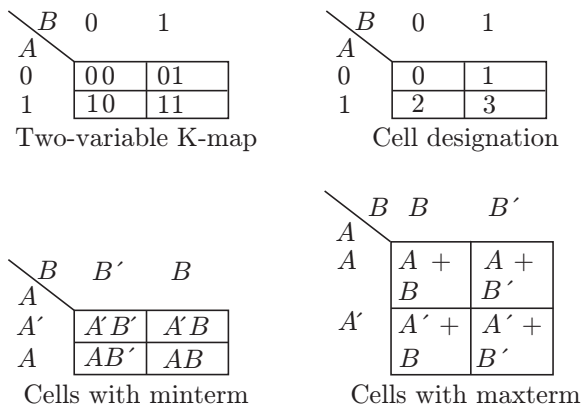


Figure 1.5 | Two-variable K-map.

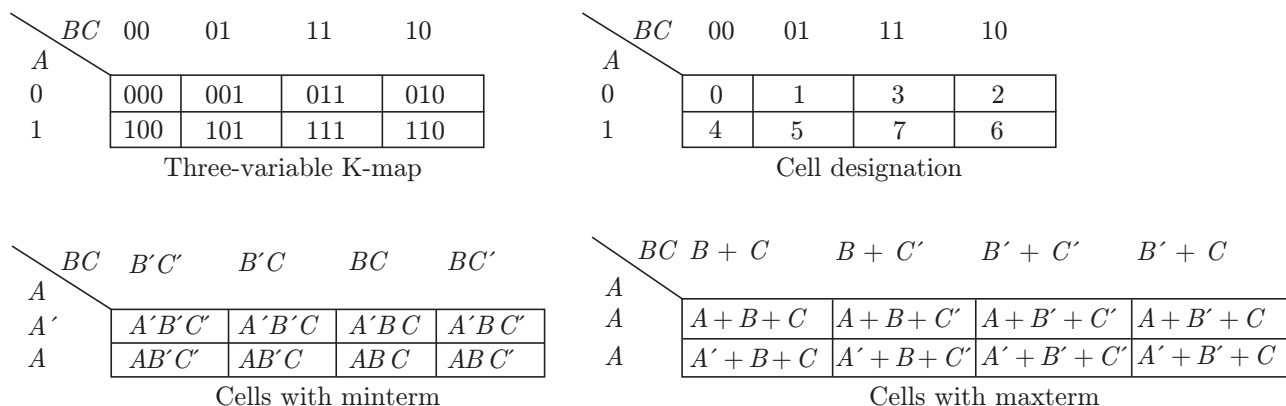


Figure 1.6 | Three-variable K-map.

Grouping of two adjacent cells removes one variable and output will contain common variables in two terms.

2. **Quads:** We can group four adjacent cells in K-map forming a quad. Grouping a quad eliminates the two variables that appear in both normal and complemented form. Resultant terms contain $(n - 2)$ variables, where n is the number of variables.
3. **Octet:** A group of eight adjacent cells is called octet. Grouping of 1's eliminates three variables. Resultant terms contain $(n - 3)$ variables, where n is the number of variables.

1.3.4.4 SOP Simplification Using K-Map

A higher priority will be given to more possible covering, that is, octet then to quad and then to pairs. The following procedure is followed:

1. Check the SOP expression and convert to standard and canonical form if required.
2. Draw the K-map based on the number of variables.
3. On the basis of SOP canonical form, 1's are entered to corresponding cells. Group the adjacent cells and take the common variable from each grouping.
4. Combine each group variable to obtain simplified expression.

1.3.4.5 POS Simplification Using K-Map

1. Check the POS expression and convert it to standard and canonical form if required.
2. Draw the K-map based on the number of variables.
3. Enter 0's in corresponding cells and group the adjacent cells.
4. Take the common variable from each grouping and write terms in OR form. The uncomplemented variable is assigned 0 and complemented is assigned 1.
5. Combine each OR term from the different groups to obtain simplified expression.

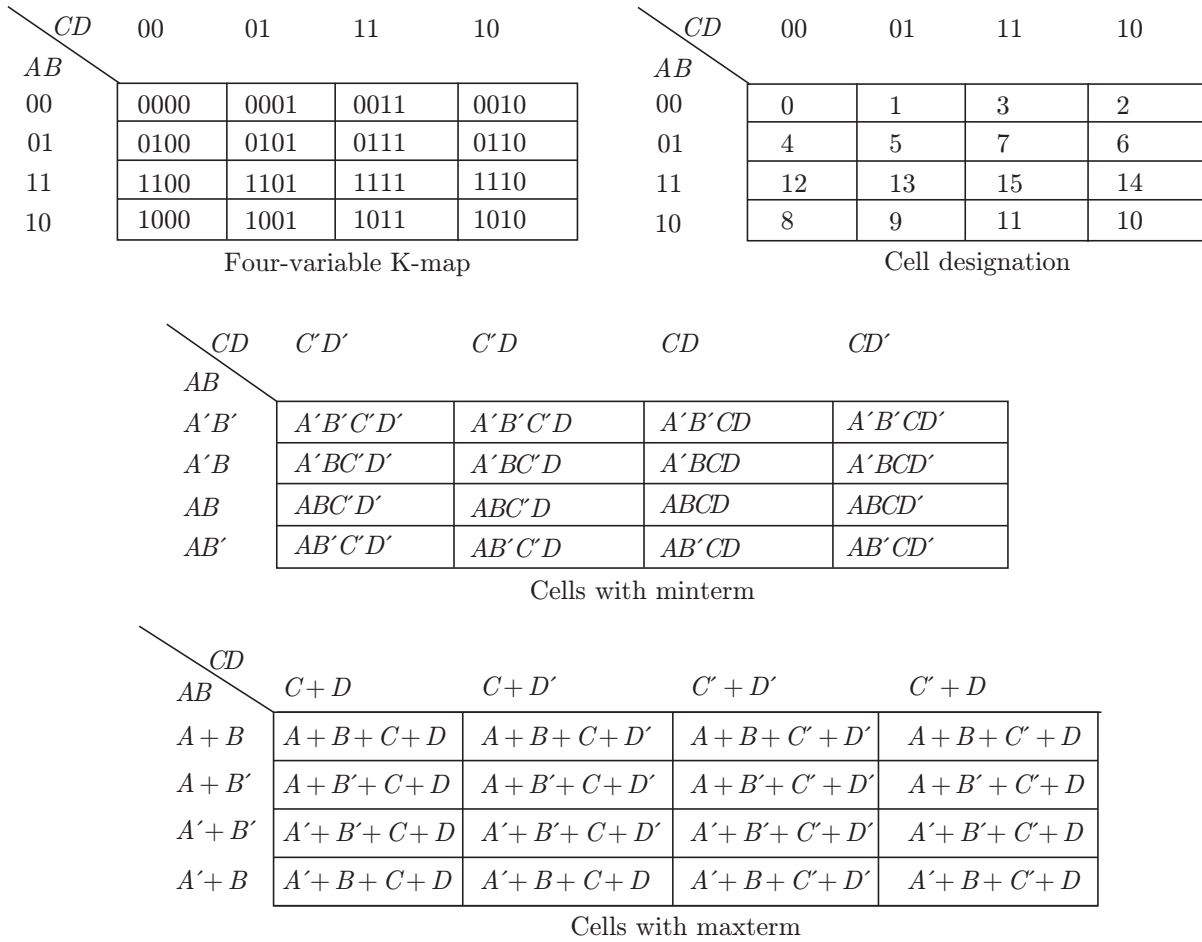


Figure 1.7 | Four-variable K-map.

Problem 1.9: Simplify $F(A, B, C, D) = \sum m(0, 2, 5, 7, 13, 15)$.

Solution:

$CD \backslash AB$	00	01	11	10	
00	1			1	$\swarrow \bar{A}\bar{B}D$
01		1	1		
11		1	1		$\swarrow BD$
10					

The simplified Boolean expression is
 $F = A'B'D' + BD$.

Problem 1.10: Simplify $F(A, B, C, D) = ACD + A'B + D'$

Solution: Converting to standard SOP form. The first term ACD has one missing variable B .

$$ACD = ACD(B + B') = ABCD + AB'CD$$

Similarly,

$$A'B = A'B(C + C')(D + D')$$

$$= A'BCD + A'BCD' + A'BC'D + A'BC'D'$$

$$D' = D'(B + B')(C + C') + (A + A')$$

Combining all terms, we get

$$F = ABCD + AB'CD + A'BCD + A'BCD'$$

$$+ A'BC'D + A'BC'D' + ABC'D'$$

$$+ AB'CD' + AB'C'D' + A'B'CD' + A'B'C'D'$$

$CD \backslash AB$	00	01	11	10
00	1			1
01	1	1	1	1
11	1		1	1
10	1		1	1

The simplified Boolean expression is $F = D' + A'B + AC$.

Problem 1.11: Simplify $S = \Pi M(0, 1, 2, 3, 4, 7)$

Solution:

$BC \backslash A$	00	01	11	10
0	0	0	0	0
1	0	1	0	
4	0		0	
5				
7				
6				

The simplified Boolean expression is $S = A \cdot (B + C) \cdot (B' + C')$

Problem 1.12: Simplify $F(A, B, C, D) = \Pi M(4, 5, 6, 7, 8, 12) \cdot d(0, 1, 2, 3, 9, 11, 14)$

Solution: A logic circuit output may be either 0 or 1 but some circuit has output that will never occur for certain combinations of inputs. These conditions are don't care conditions and represented by X in K-map.

$CD \backslash AB$	00	01	11	10
00	X	X	X	X
01	O	O	O	O
11	O			X
10	O	X	X	

The simplified Boolean expression is $F = A(C + D)$.

1.4 DIGITAL CIRCUITS

The classification of digital circuits is shown in Fig. 1.8.

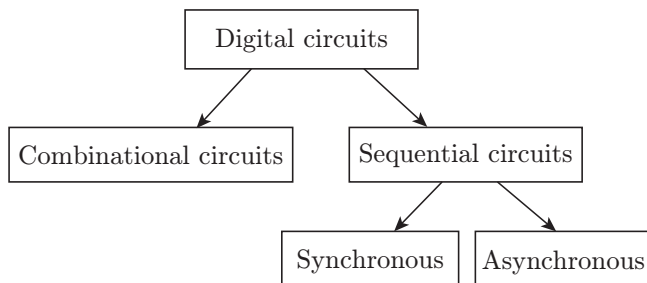


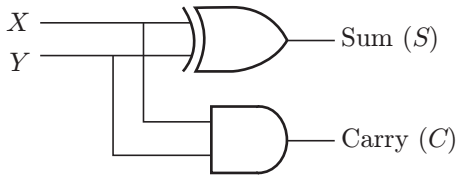
Figure 1.8 | Types of digital circuits.

1.4.1 Combinational Circuits

A logic circuit whose output at any instant of time depends only on the present input is called combinational circuit. It contains no memory element.

1.4.1.1 Half Adder

A half adder is a combinational circuit for the addition of two 1-bit numbers. X and Y are inputs and outputs are sum (S) and carry (C) (Fig. 1.9 and Table 1.4).

**Figure 1.9** | Logic diagram of a half adder.**Table 1.4** | Truth table

Inputs		Outputs	
<i>X</i>	<i>Y</i>	Carry (<i>C</i>)	Sum (<i>S</i>)
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

The sum and carry are as follows:

$$S = X'Y + XY'$$

$$C = XY$$

1.4.1.2 Full Adder

A full adder is a combinational circuit that performs the addition of 3 bits (2 significant bits and previous carry). It consists of three inputs and two outputs, two input bits that are to be added, the third input is the carry from the previous position (Fig. 1.10 and Table 1.5).

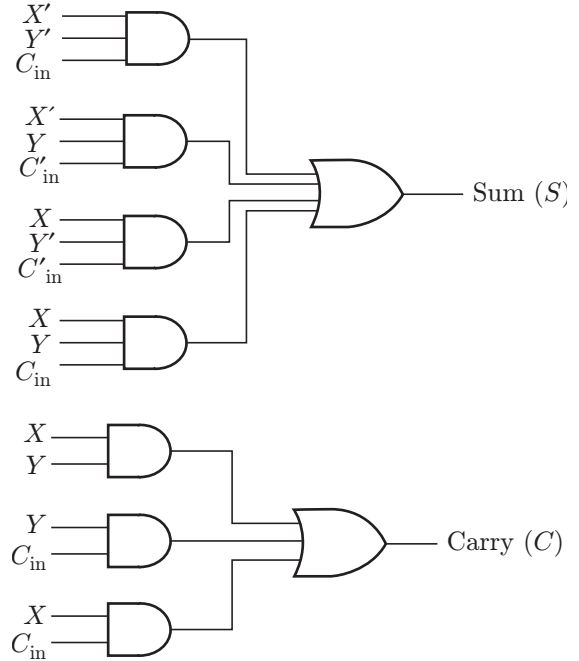
Table 1.5 | Truth table

Inputs			Outputs	
<i>X</i>	<i>Y</i>	<i>C_{in}</i>	Sum (<i>S</i>)	<i>C_{out}</i>
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The sum and carry are as follows:

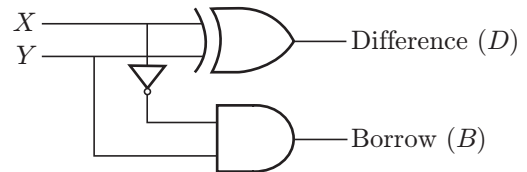
$$S = X'Y'C_{in} + X'YC'_{in} + XYC_{in}$$

$$C_{out} = AC_{in} + XY + YC_{in}$$

**Figure 1.10** | Logic diagram of a full adder.

1.4.1.3 Half Subtractor

A half subtractor is a combinational circuit that subtracts 2 bits (minuend and subtrahend) and produces their differences and borrow as output (Fig. 1.11 and Table 1.6).

**Figure 1.11** | Logic diagram of a half subtractor.**Table 1.6** | Truth table

Inputs		Outputs	
<i>X</i>	<i>Y</i>	Difference (<i>D</i>)	Borrow (<i>B</i>)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

The difference and borrow are as follows:

$$D = XY' + X'Y$$

$$B = XY$$

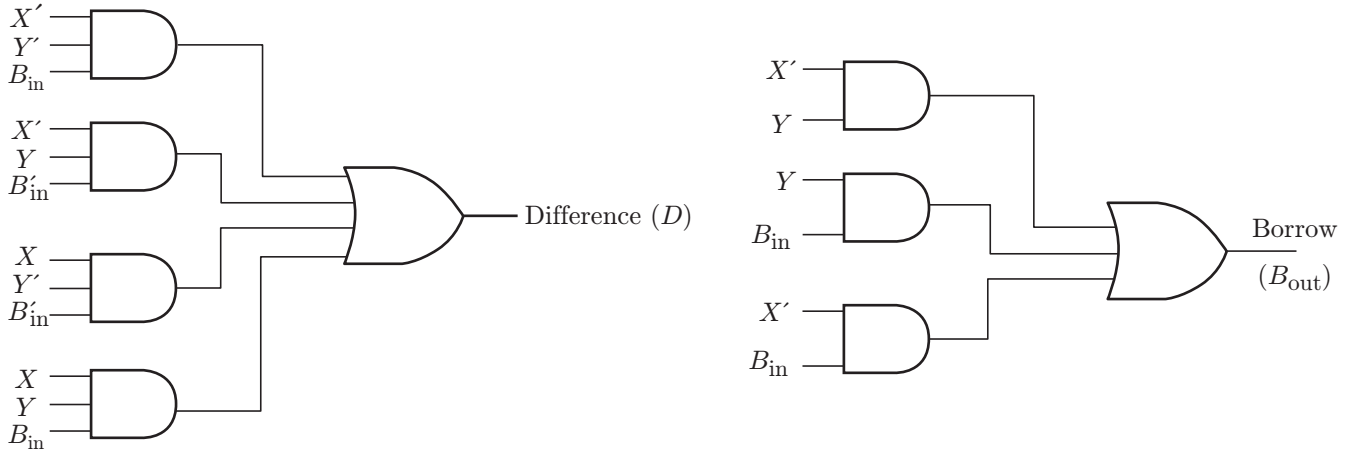


Figure 1.12 | Logic diagram of a full subtractor.

1.4.1.4 Full Subtractor

A full subtractor is a combinational circuit that performs subtraction involving 3 bits (minuend bit, subtrahend bit and borrow from previous stage) (Table 1.7 and Fig. 1.12).

Table 1.7 | Truth table

Inputs			Outputs	
X	Y	B_{in}	Difference (D)	B_{out}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

The difference and borrow are as follows:

$$D = X'Y'B_{in} + X'YB'_{in} + XY'B_{in} + XYB_{in}$$

$$B = X'Y + X'YB_{in} + YB_{in}$$

1.4.1.5 Parallel Adder (Ripple Carry Adder)

A parallel adder consists of full adders connected in cascade, that is, carry output of each adder is connected to the carry input of the next higher-order adder (Fig. 1.13). An n -bit parallel adder is constructed using $(n - 1)$ full adders and 1 half adder. As full adder consists of 2 half

adders and 1 OR gate, so n -bit parallel adder can be made from $(2n - 1)$ half adders and $(n - 1)$ OR gates.

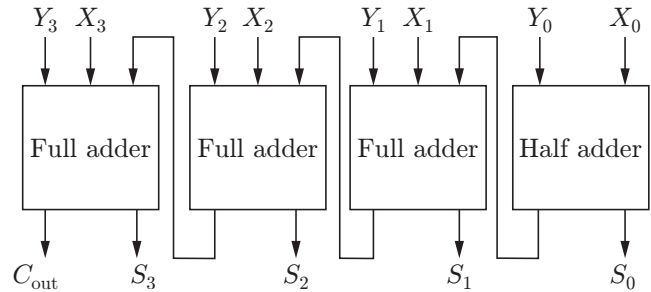


Figure 1.13 | Block diagram of a parallel adder.

1.4.1.6 Look-Ahead Carry Header

In a parallel adder, carry propagation delay is present. To overcome this difficulty, look-ahead carry adder is used. It uses logic gates to look at lower-order bits of augend and addend to check if higher-order carry is generated or not. It uses two functions, carry generate and carry propagate, for the same (Fig. 1.14).

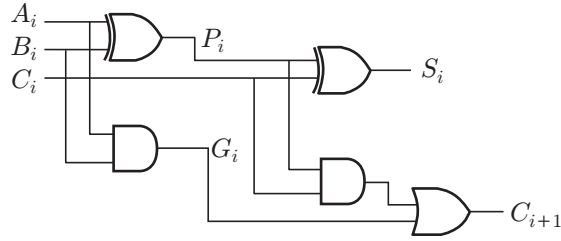
Problem 1.13: How many AND and OR gates are required to design 7-bit look-ahead carry adder.

Solution: To design a 7-bit look-ahead carry adder, we need

$$\text{AND gate} = \frac{n(n+1)}{2} = \frac{7(7+1)}{2} = 28$$

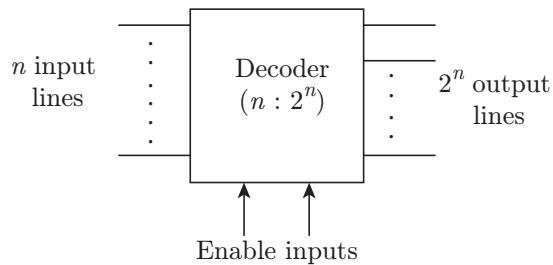
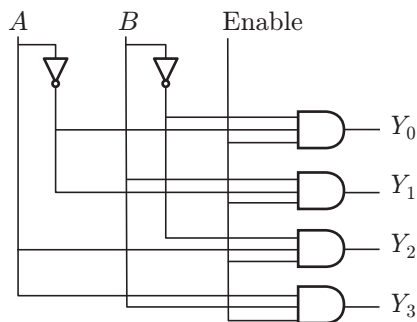
$$\text{OR gate} = n = 7$$

where n is the number of bits.

Full Adder circuit with P_i and G_i Carry propagation $P_i = A_i \oplus B_i$ Carry generation $G_i = A_i B_i$ Output sum $S_i = P_i \oplus C_i$ Output carry $C_{i+1} = G_i + P_i C_i$ **Figure 1.14** | Logic diagram of a look-ahead adder.

1.4.1.7 Decoder

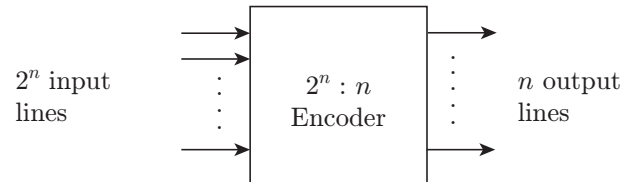
A decoder is a combinational circuit that converts binary information from n input lines to a maximum of 2^n unique output lines (Fig. 1.15). The size of a decoder is n to m where $m \leq 2^n$. It is used to route input data to a specified output line for memory addressing (Fig. 1.16 and Table 1.8).

**Figure 1.15** | Block diagram of a decoder.**Figure 1.16** | Logic diagram of a 2X4 decoder.**Table 1.8** | Truth table of 2X4 decoder

Inputs		Outputs			
A	B	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

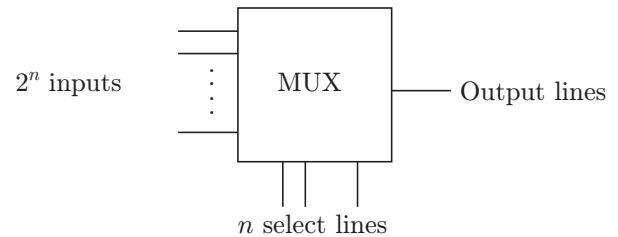
1.4.1.8 Encoder

An encoder is a combinational circuit that has 2^n (or fewer) inputs and n output lines (Fig. 1.17). It accepts an active level on one of its inputs representing a decimal or octal digit and converts it to a coded output.

**Figure 1.17** | Block diagram of an encoder.

1.4.1.9 Multiplexer

A multiplexer is a combinational circuit that provides inputs one by one at single output; and which input will be out at any instant depends upon the combination at select lines (Fig. 1.18). It converts parallel data to serial data (Fig. 1.19 and Table 1.9).

**Figure 1.18** | Block diagram of a multiplexer.**Table 1.9** | Truth table of 4 to 1 MUX

Select Lines		Outputs
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

The Boolean function is

$$Y = I_0 S_0' S_1' + I_1 S_0 S_1' + I_0 S_0' S_1 + I_0 S_0 S_1$$

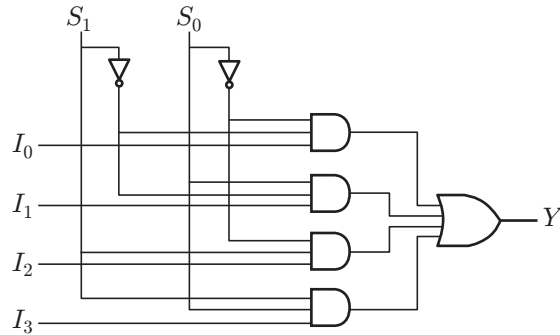


Figure 1.19 | Logic diagram of 4 to 1 MUX.

1.4.1.10 Boolean Function Implementation Using Multiplexers

1. Take one variable for input lines and others for selection lines.
2. Write the minterms with the variable in complemented form in the first row and with variable selected in uncomplemented form in the second row.
3. Then encircle the minterms which are present in the function.
 - If there is circled variable in the column, then we put 0 on the corresponding line.
 - If there are circled variables, then we put 1 on the line.
 - If bottom variable is circled and top is not circled, A is used in input line.
 - If bottom variable is not circled and top is circled, A' is used as input to that line.

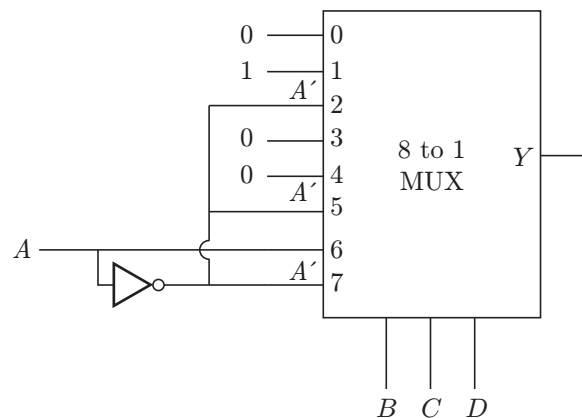
Problem 1.14: Implement the function $F(A, B, C, D) = \Sigma m(1, 2, 5, 7, 9, 14)$ using MUX.

Solution: Select variable A for input as B, C and D for selection lines.

As there are four variables so size of MUX is $2^{N-1} = 2^3 = 8$ to 1 MUX.

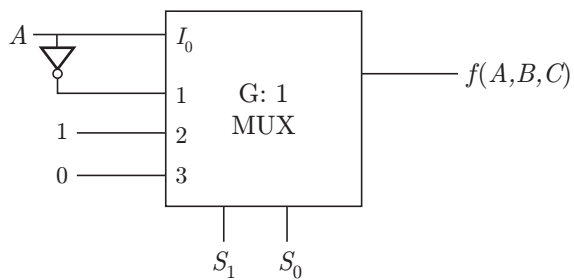
1. Minterms with A' are 0 – 7.
2. Minterms with A are 8 – 15.

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	0	①	②	3	4	⑤	6	⑦
A	8	⑨	10	11	12	13	⑭	15
	0	1	A'	0	0	A'	A	A'



Logic diagram

Problem 1.15: The combinatorial circuit below is equivalent to _____.



Solution:

	I_0	I_1	I_2	I_3
A'	0	①	②	3
A	④	5	⑥	7
	A	A'	1	0

It is equivalent to $\sum m(1, 2, 4, 6)$.

1.4.1.11 Demultiplexer

A demultiplexer is a combinational circuit that receives information on a single line and transmits this on one of 2^n of possible output line (Fig. 1.20). The selection of possible output line is controlled by bit values of n select lines.

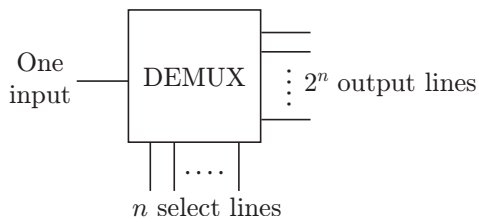


Figure 1.20 | Block diagram of a demultiplexer.

Implementation of higher order decoders and multiplexers by using lower order. Whenever odd number implementation is there, one NOT gate is required.

Example 1.3

Design 4×16 decoders using 2×4 decoders

$$16/4 = 4$$

$$4/4 = 1$$

$$\text{Total} = 5$$

Example 1.4

Design 8×256 decoders using 4×64 decoders

$$256/64 = 4$$

$$4/64 = 0$$

$$\text{Total} = 4$$

Example 1.5

Design 6×64 decoders using 2×4 decoders

$$64/4 = 16$$

$$16/4 = 4$$

$$4/4 = 1$$

$$\text{Total} = 21$$

1.4.2 Sequential Circuits

A logic circuit whose output at any instant of time depends on the present input as well as past output is known as sequential circuit. It contains memory element.

1.4.2.1 Flip-Flops

A flip-flop or bistable multivibrator is a synchronous bistable circuit made up of logic gates that can exist in either of two stable states in presence of a falling or raising edge of a clock signal. Flip-flops can change its state by means of some external signal. It can “store” binary information. There are different types of flip-flops based on the manner the inputs affect their output state:

1. *RS* flip-flop
2. *D* flip-flop
3. *JK* flip-flop
4. *T* flip-flop

RS Flip-Flop (Direct Coupled)

This sequential logic circuit has two stable states ($Q = 1$ set state and $Q = 0$ reset state) achieved by giving proper inputs to R and S inputs (Figs. 1.21 and 1.22 and Table 1.10). If the circuit is in one particular state it continues to remain in that state and can store 1-bit data.

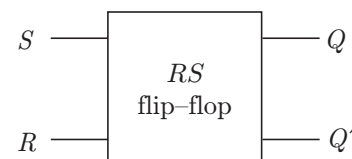


Figure 1.21 | Block diagram of an *RS* flip-flop.

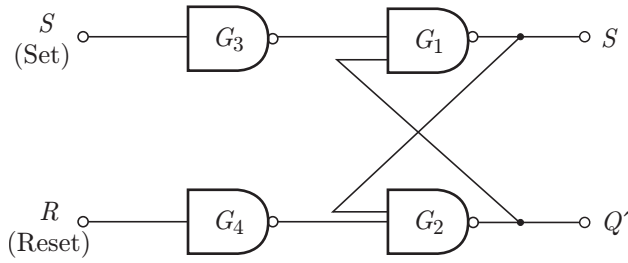


Figure 1.22 | Logic diagram of an *RS* flip-flop.

Table 1.10 | Truth table of *RS* flip-flop

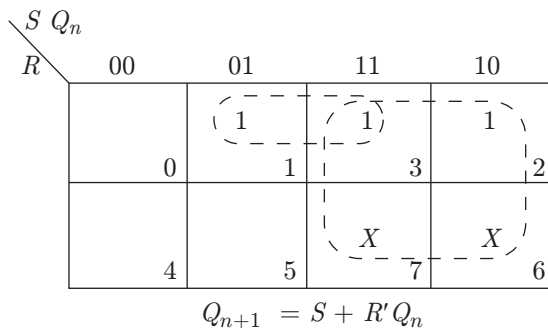
Inputs		Outputs
S_n	R_n	Q_{n+1}
0	0	Q_n (No change)
0	1	0 (Reset)
1	0	1 (Set)
1	1	Prohibited

The characteristic equation which gives the algebraic description of the next state of a flip-flop obtained by K-map simplification is as follows (see Table 1.11):

$$Q_{n+1} = S + R'Q_n$$

Table 1.11 | Truth table

Flip-Flop Inputs		Present State	Next State
R_n	S_n	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	X
1	1	1	X



Clocked *RS* Flip-Flop

This circuit sets or resets the memory cell in synchronization with the clock pulse (Figs. 1.23 and 1.24 and Table 1.12). If clock pulse is not present, the gates G_3 and G_4 are inhibited; their outputs are 1.

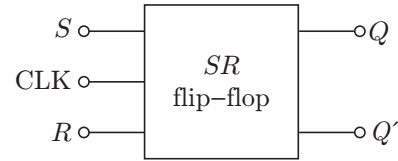


Figure 1.23 | Block diagram of a clocked *SR* flip-flop.

Table 1.12 | Truth table of *SR* flip-flop

Clock	Flip-Flop Inputs		Outputs
CLK	S_n	R_n	Q_{n+1}
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Prohibited
0	X	X	Q_n

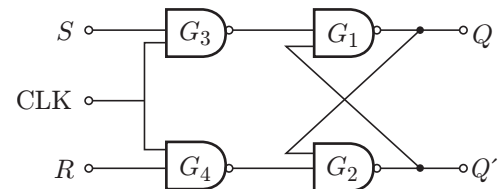
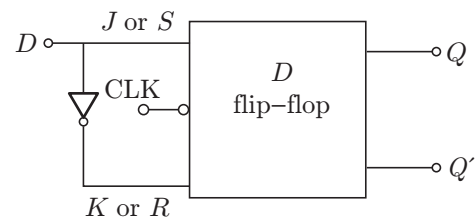


Figure 1.24 | Logic diagram of a clocked *SR* flip-flop.

D Flip-Flop

This circuit has only one input terminal, and output is followed by input. It has delay of exactly one clock cycle (Figs. 1.25 and 1.26 and Table 1.13). The input to R is through an inverter from S .

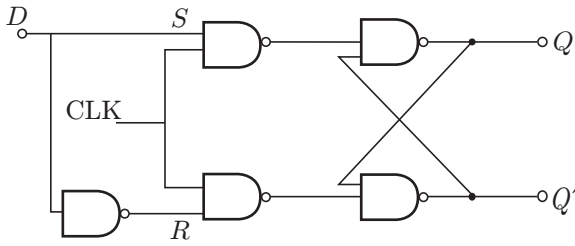


JK or *SR* Flip-Flop converted to *D*

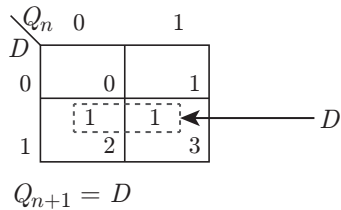
Figure 1.25 | Block diagram of a *D* flip-flop.

Table 1.13 | Truth table of D flip-flop

Flip-Flop Inputs	Present State	Next State
D_n	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

**Figure 1.26** | Logic diagram of a D flip-flop.

The characteristic equation is $Q_{n+1} = D$ (Fig. 1.27)

**Figure 1.27** | K-Map

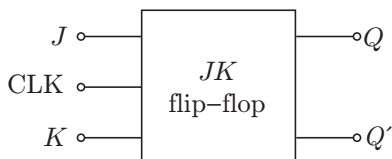
JK Flip-Flop

The uncertainty in SR flip-flop when $S_n = R_n = 1$ can be avoided by using JK flip-flop (Figs. 1.28 and 1.29 and Table 1.14). The output of JK flip-flop is connected back to the inputs of gate. The inputs J and K are ANDed with Q and Q' to obtain S and R inputs:

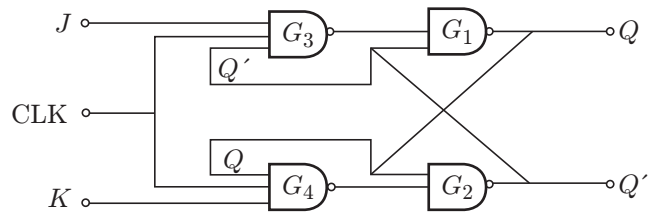
$$S = JQ'$$

$$R = J \cdot Q$$

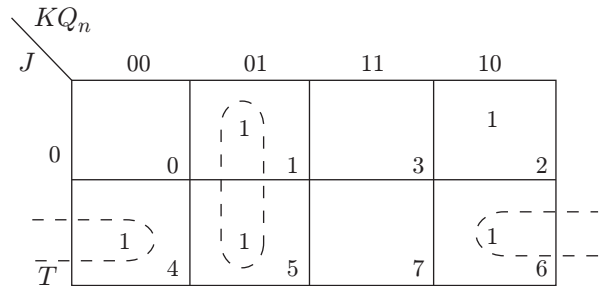
The flip-flop sets when input J is high and resets when input K is high. If $J = K = 1$, the flip-flop will toggle on the next positive clock.

**Figure 1.28** | Block diagram of a JK flip-flop.**Table 1.14** | Truth table of JK flip-flop

Flip–Flop Inputs		Present State	Next State
J_n	K_n	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

**Figure 1.29** | Logic diagram of a JK flip-flop.

The characteristic equation is $Q_{n+1} = JQ'_n + KQ_n$ (Fig. 1.30).



$$Q_{n+1} = K'Q_n + JQ'_n$$

Figure 1.30 | K-Map

Repetition of toggle for a single-clock pulse is known as **race-around problem**. The output of JK flip-flop will oscillate between 0 and 1 during a single-clock pulse so that at the end of clock pulse the value of Q is uncertain. This problem can be avoided by decreasing the pulse width, by edge triggering or increasing propagation delay.

Master-Slave JK Flip-flop

It is a cascade of two RS flip-flops (master and slave) with feedback from outputs of slave to input of master to avoid race-around problem. Positive clock pulses are applied to first flip-flop and pulses are inverted before applying to slave flip-flop (Fig. 1.31).

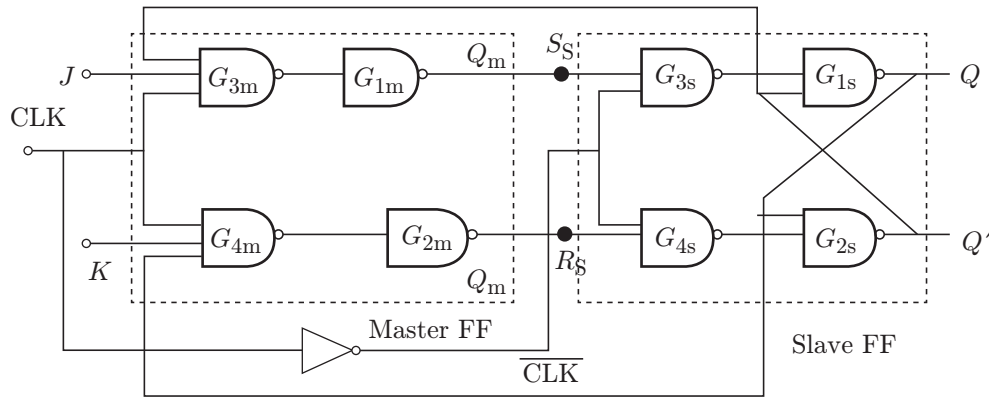


Figure 1.31 | Block diagram of a master–slave JK flip–flop.

T Flip–flop

It is also known as toggle flip–flop, which is obtained by connecting both inputs of JK flip–flop (Fig. 1.32 and Table 1.15).

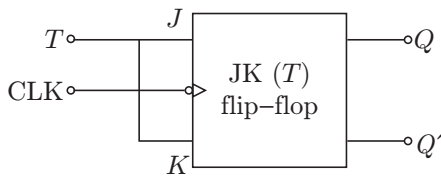


Figure 1.32 | Block diagram of a T flip–flop.

Table 1.15 | Truth table of T flip–flop

Flip–Flop Inputs		Present State	Next State
T_n		Q_n	Q_{n+1}
0		0	0
0		1	1
1		0	1
1		1	0

Q_n	0	1
T		
0	0	1
1	1	3

$$Q_{n+1} = T'Q_n + TQ_n'$$

Figure 1.33 | K-Map

Excitation Table of Flip–Flop

The excitation table of flip–flop states is given in Table 1.16.

Conversion Equation

The flip–flop conversion equations are given in Table 1.17.

Table 1.16 | Excitation Table of Flip–flop States

Flip–Flop States		RS Flip–Flop Inputs		JK Flip–Flop Inputs		D Flip–Flop Inputs	T Flip–Flop Inputs
Present State (Q_n)	Next State (Q_{n+1})	R	S	J	K		
0	0	X	0	0	X	0	0
0	1	0	1	1	X	1	1
1	0	1	0	X	1	0	1
1	1	0	X	X	0	1	0

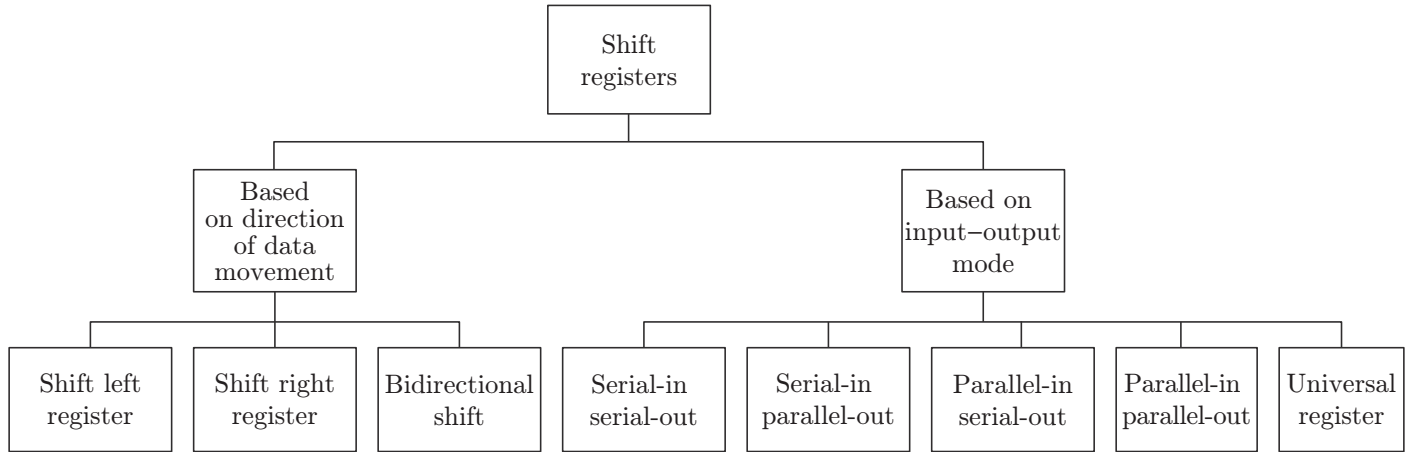


Figure 1.34 | Types of registers

Table 1.17 | Flip-flop conversion equations

1	SR to JK	$S = JQ'$ $R = K \cdot Q$
2	SR to D	$S = D$ $R = D'$
3	SR to T	$S = TQ'$ $R = TQ$
4	JK to SR	$J = S$ $K = R$
5	JK to D	$J = D$ $K = D'$
6	JK to T	$J = T$ $K = T$
7	D to JK	$D = TQ' + K'Q$
8	D to SR	$D = S + R'Q$
9	D to T	$D = T \oplus Q$
10	T to SR	$T = SQ' + RQ$
11	T to D	$T = D \oplus Q$
12	T to JK	$T = JQ' + KQ$

Applications of Flip-Flops

Registers

Registers are collection of flip-flops, and each flip-flop is capable of storing single bit of information. They make stored information available to the logic elements for the

computation. There are different types of registers as shown in Fig. 1.34.

1. Serial-In/Serial-Out Shift Register: It accepts data serially on single input line and produces output in serial form. Data can be shifted right or left.

- **Shift right register:** In shift right register, serial data is entered from the left side of a register and leaves from the right side. For example, a 4-bit number 1101 ($Q_D Q_C Q_B Q_A$) is entered serially into first flip-flop D from the left side (Fig. 1.35).

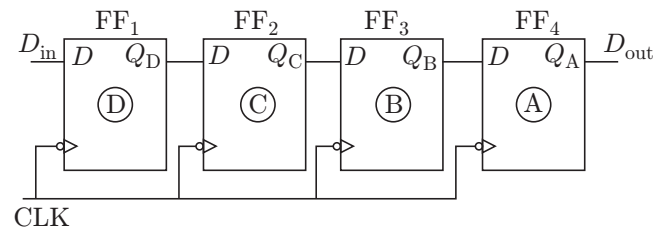


Figure 1.35 | 4-Bit shift right register.

The content of register after each clock pulse is shown in Table 1.18.

Table 1.18 | Content of register after each clock pulse

Clock Pulse	Q_D	Q_C	Q_B	Q_A
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	1	0	1	0
4	1	1	0	1

- **Shift left register:** In shift left register, serial data is entered from the right side of the register and leaves from the left side. For example, a 4-bit number 1101 ($Q_D Q_C Q_B Q_A$) is entered serially into input of shift left register (Fig. 1.36).

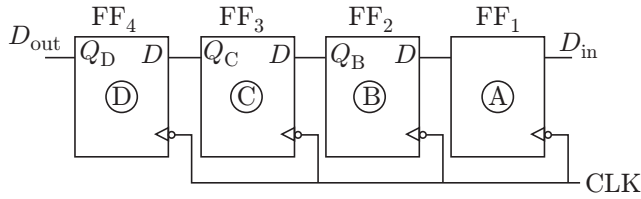


Figure 1.36 | 4-Bit shift left register.

The content of register after each clock pulse is given in Table 1.19.

Table 1.19 | Content of register after each clock pulse

Clock Pulse	Q_D	Q_C	Q_B	Q_A
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	1	1	0
4	1	1	0	1

Note: A universal register can perform all the operations of shift register.

- 2. Serial-In to Parallel-Out (SIPO):** The register is loaded with data serially, one bit at a time, with the stored data being available at the output in parallel form (Fig. 1.37).

The content of register after each clock pulse is given in Table 1.20.

Table 1.20 | Content of register after each clock pulse

Clock Pulse	Q_A	Q_B	Q_C	Q_D
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	0	0	0	0

- 3. Parallel-In to Serial-Out (PISO):** The parallel data is loaded simultaneously into the register and is shifted out of the register serially, one bit at a time (Fig. 1.38).

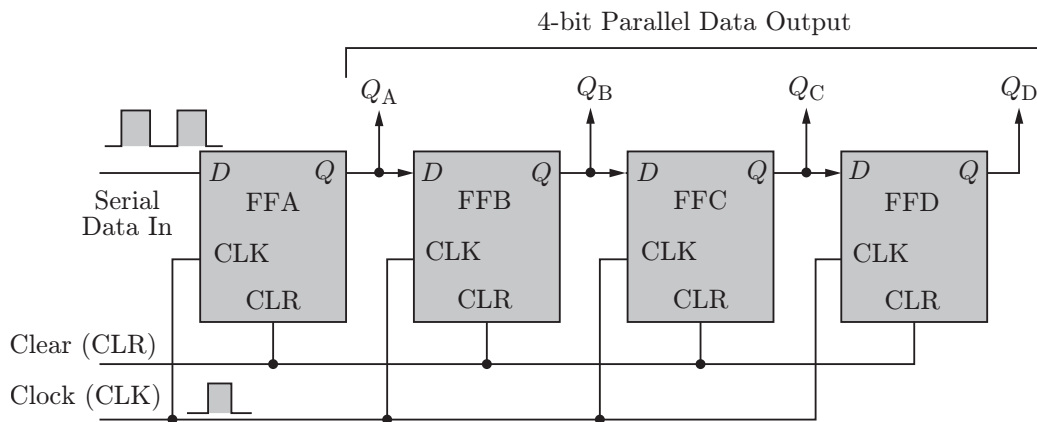


Figure 1.37 | 4-Bit SIPO register.

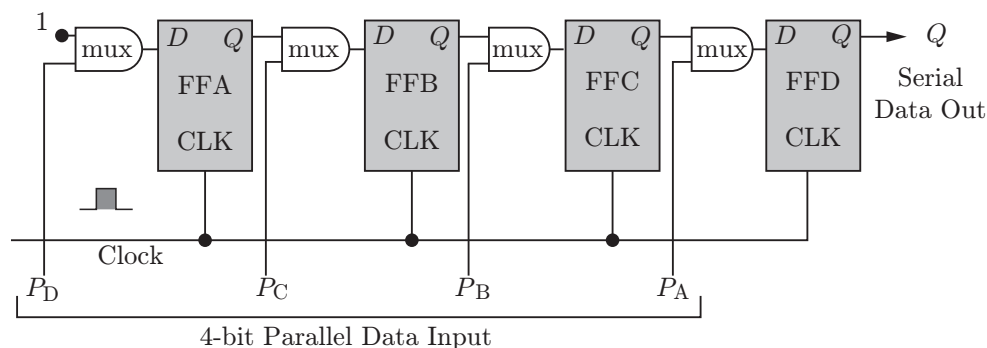


Figure 1.38 | 4-Bit PISO register.

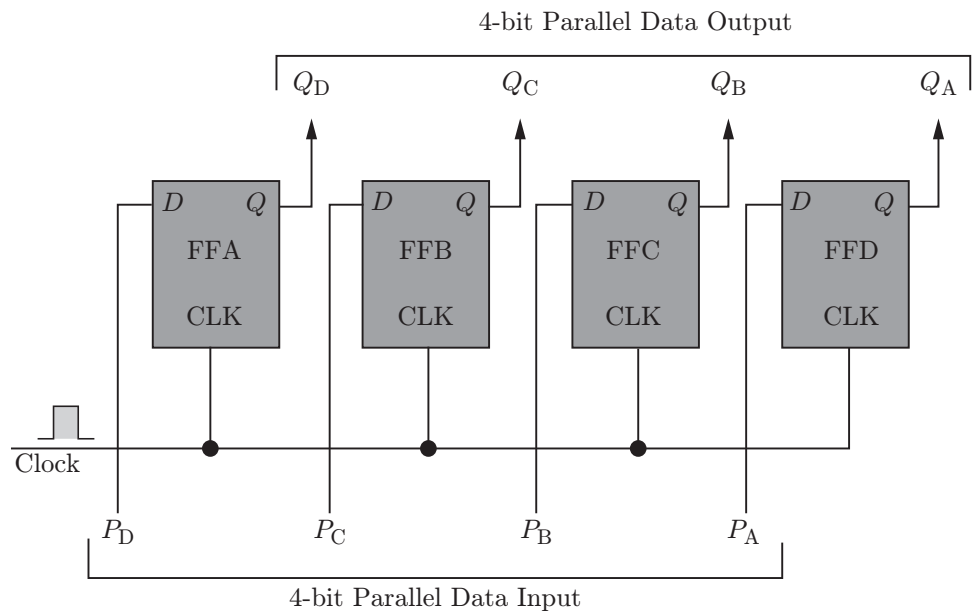


Figure 1.39 | 4-Bit PIPO register.

4. Parallel-In to Parallel-Out (PIPO): The parallel data is loaded simultaneously into the register, and transferred together to their respective outputs by the same clock pulse (Fig. 1.39).

Counters

A counter is a sequential logic circuit that is capable of counting the number of clock pulses arriving at its input in a cyclic sequence. It is just a register that shifts through a predetermined sequence of states upon the application of input clock pulses. There are two different types of counters – synchronous counter (parallel counter) and asynchronous counter (serial or ripple counter) (Table 1.21).

Table 1.21 | Differences between asynchronous and synchronous counter.

Asynchronous Counter	Synchronous Counter
The output of first flip-flop drives the clock for next flip-flop. So all flip-flops are not clocked simultaneously.	There is no connection between output of first flip-flop and clock input of next flip-flop. All the flip-flops are clocked simultaneously.
Speed is low.	Speed is high.
Simple logic circuit as number of states increases.	Complex design as number of states increases.
Example: Ring counter, Johnson counter.	Example: Binary ripple counter, Up-down counter.

The counter with n flip-flops has a maximum modulus 2^n so it can count from 0 to $2^n - 1$. It can also produce MOD numbers less than 2^n by skipping the states.

1. 4-Bit Asynchronous Up Counter: Figures 1.40–1.42 show a 4-bit asynchronous counter constructed using JK flip-flops. The clock input of first flip-flop (the one with the Q_0 output) is connected to the external clock. The clock of second flip-flop drives from output Q_0 of the first flip-flop and so on. Each flip-flop input ($J = K = 1$) is connected to HIGH voltage so it toggles when negative edge is arrived at clock input and counts from 0000 to 1111.

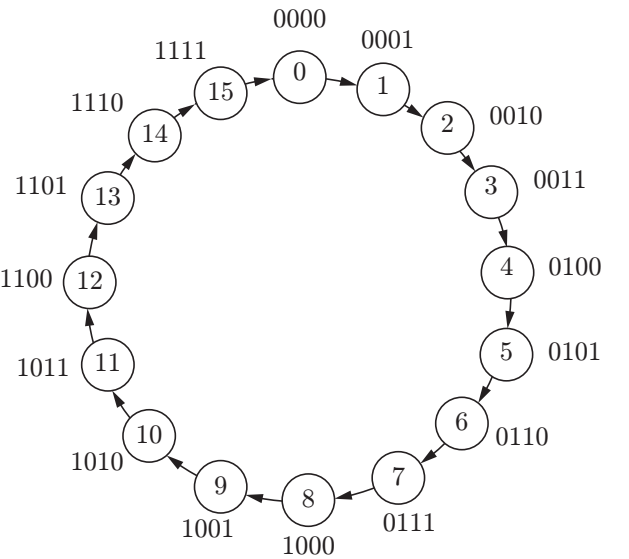


Figure 1.40 | State diagram of a 4-bit asynchronous counter.

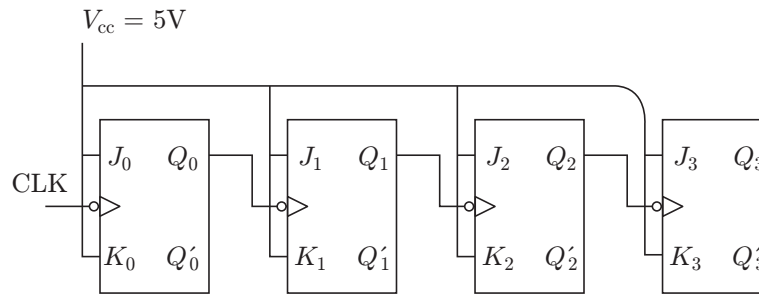


Figure 1.41 | Logic diagram for a 4-bit asynchronous counter.

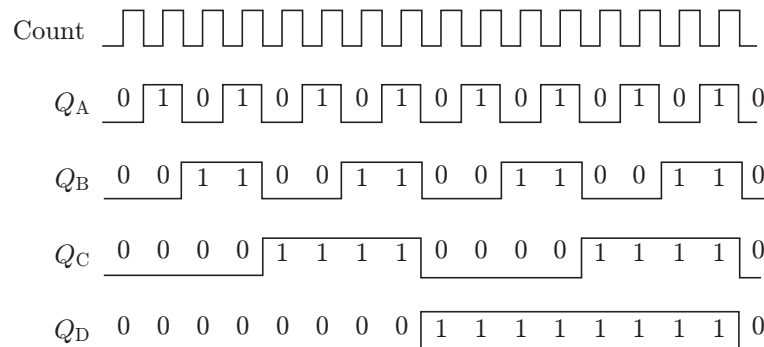


Figure 1.42 | Timing diagram for a 4-bit asynchronous counter.

2. MOD-10 Asynchronous Counter: A MOD-10 asynchronous counter has 10 states and number of flip-flops required are $2^4 \geq 10$ (Figs. 1.43 and 1.44). It counts the sequence from 0000 to 1001 and after that it returns to the initial state.

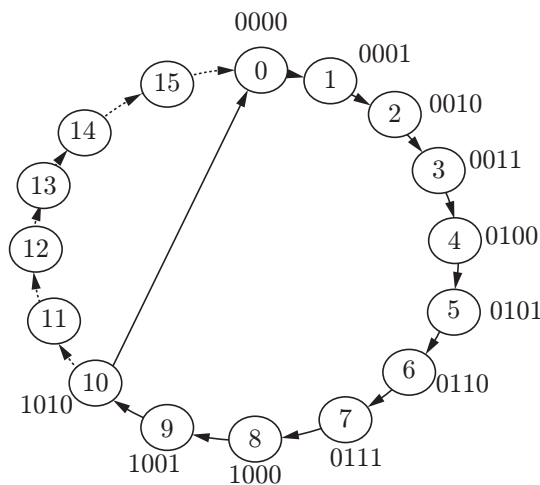


Figure 1.43 | State diagram of a 4-bit MOD-10 counter.

3. 4-Bit Synchronous Counter: In a 4-bit synchronous counter, the least significant bit of flip-flop is connected at high level and other inputs of JK flip-flop are driven by some combination of flip-flop outputs. Figure 1.45 shows the logic diagram of a 4-bit synchronous counter. The output of flip-flop0 always toggle, and that of flip-flop1 toggles when Q_0 is 1,

otherwise it maintains the previous state. The flip-flop 2 toggles when both Q_1 and Q_0 are 1. The flip-flop 3 toggles when Q_2 , Q_1 and Q_0 are 1 (Table 1.22).

Table 1.22 | Truth table for 4-bit synchronous counter

Count	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	0	0
15	1	1	0	1

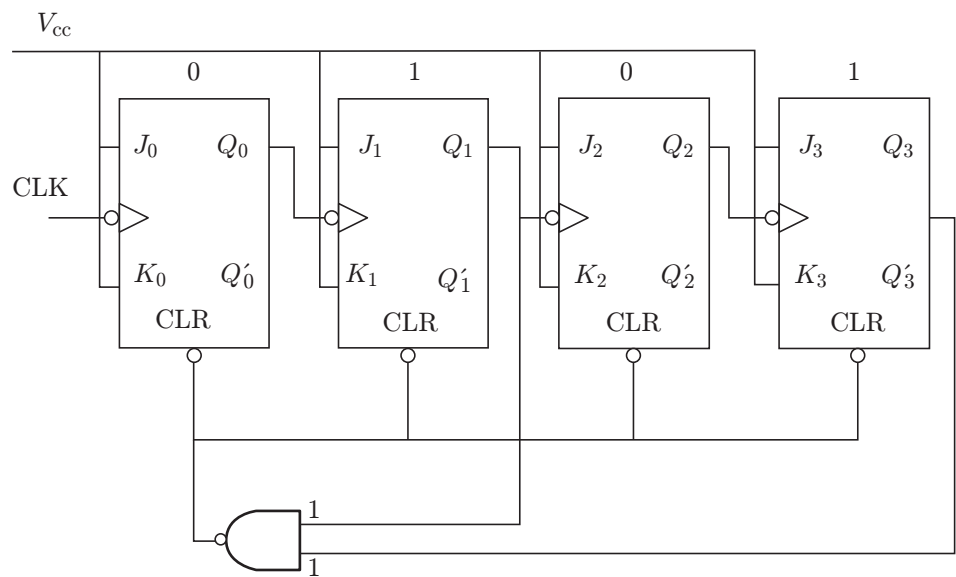


Figure 1.44 | Logic diagram for a MOD-10 asynchronous counter.

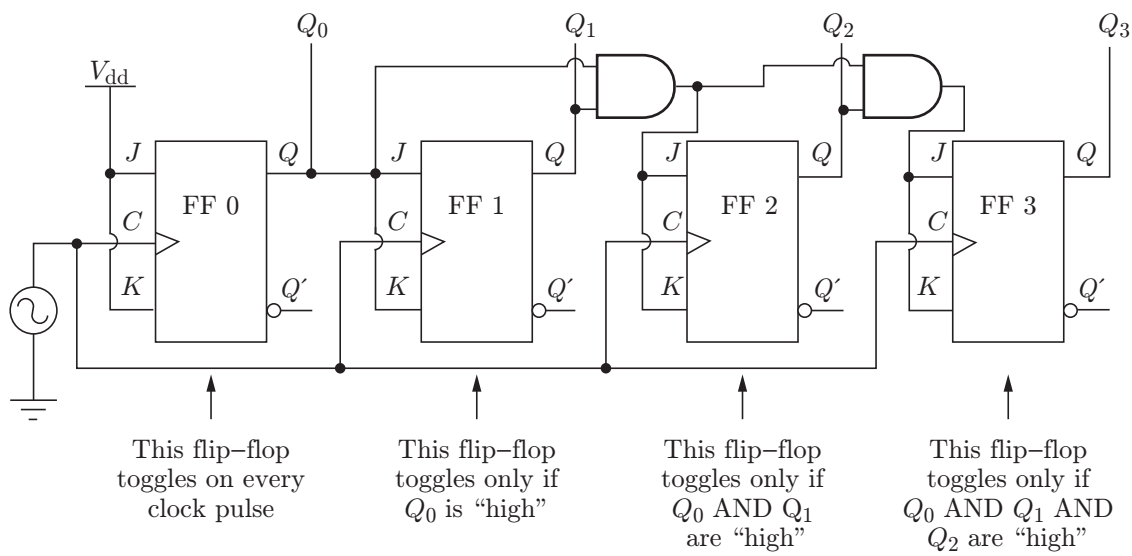


Figure 1.45 | Logic diagram of a 4-bit synchronous counter.

4. MOD-5 Synchronous Counter:

Step 1: Number of states is five, so counting sequence 0 to 4.

Flip-flops required are $2^n \geq 5$, so $n = 3$.

Let us label these flip-flops as A, B and C.

Step 2: Obtain the excitation table for JK flip-flop (Table 1.23).

Step 3: Develop state table using the excitation table (Table 1.24).

Step 4: K-map simplification for finding expressions corresponding to each input (Fig. 1.46).

Table 1.23 | Excitation table for JK flip-flop

Flip-Flop States		JK Flip-Flop Inputs	
Present State (Q_n)	Next State (Q_{n+1})	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Table 1.24 | State table

Present State			Next State			Flip-Flop Inputs					
Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	J_C	K_C	J_B	K_B	J_A	K_A
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	0	0	0	X	1	0	X	0	X
1	0	1	X	X	X	X	X	X	X	X	X
1	1	0	X	X	X	X	X	X	X	X	X
1	1	1	X	X	X	X	X	X	X	X	X

Expression for J_C

$Q_B Q_A$	00	01	11	10
Q_C				
0			1	
1	X	X	X	X

$$J_C = Q_B Q_A$$

Expression for K_C

$Q_B Q_A$	00	01	11	10
Q_C				
0	X	X	X	X
1	X	X	X	X

$$K_C = 1$$

Expression for J_B

$Q_B Q_A$	00	01	11	10
Q_C				
0		1	X	X
1		X	X	X

$$J_B = Q_B$$

Expression for K_B

$Q_B Q_A$	00	01	11	10
Q_C				
0	X	X	1	
1	X	X	X	X

$$K_B = Q_A$$

Expression for J_A

$Q_B Q_A$	00	01	11	10
Q_C				
0	1	X	X	1
1		X	X	X

$$J_A = Q'_C$$

Expression for K_A

$Q_B Q_A$	00	01	11	10
Q_C				
0	X	1	1	X
1	X	X	X	X

$$K_A = 1$$

Figure 1.46 | K-map simplification.

Step 5: Logic diagram of MOD-5 counter (Fig. 1.47).

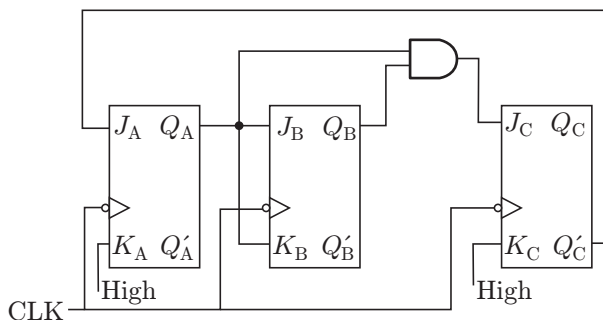


Figure 1.47 | Logic diagram of a MOD-5 counter.

5. Shift register counter: It is basically a shift register whose serial output is connected back to serial input in order to produce special sequences.

6. Ring counter: A ring counter is a circular shift register in which output of last flip-flop is connected to the input of the first in a ring (Fig. 1.48 and Table 1.25). In this, only one flip-flop is in state one while others are in their zero states. Each state repeats after every n clock cycles, where n is the number of flip-flops used. Initially the counter is reset by applying low signal to active low CLR input. All flip-flops reset except the first flip-flop.

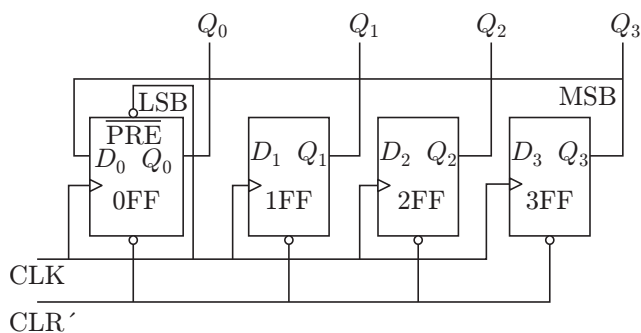


Figure 1.48 | Logic diagram of a 4-bit ring counter.

Table 1.25 | Truth table of 4-bit ring counter

CLK	CLR'	Q_3 (MSB)	Q_2	Q_1	Q_0
0	0	0	0	0	1
1	1	0	0	1	0
1	1	0	1	0	0
1	1	1	0	0	0

(Continued)

Table 1.25 | Continued

CLK	CLR'	Q_3 (MSB)	Q_2	Q_1	Q_0
1	1	0	0	0	1
1	1	0	0	1	0
1	1	0	1	0	0
1	1	1	0	0	0

7. Johnson counter: A Johnson counter (switch-tail ring counter, twisted-ring counter or walking-ring counter) is a variation of shift register counter, where the complement of the output of the last flip-flop is fed back as input to the first flip-flop (Fig. 1.49 and Table 1.26). The counter has counting sequence length of twice the length of the shift register, that is, number of flip-flops.

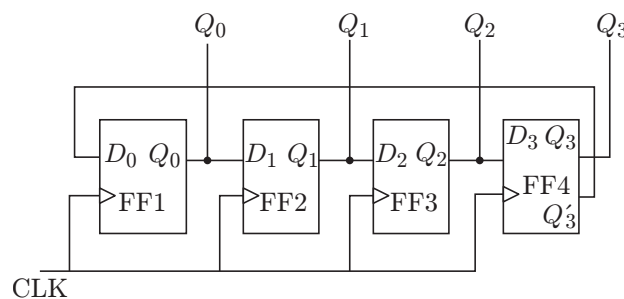


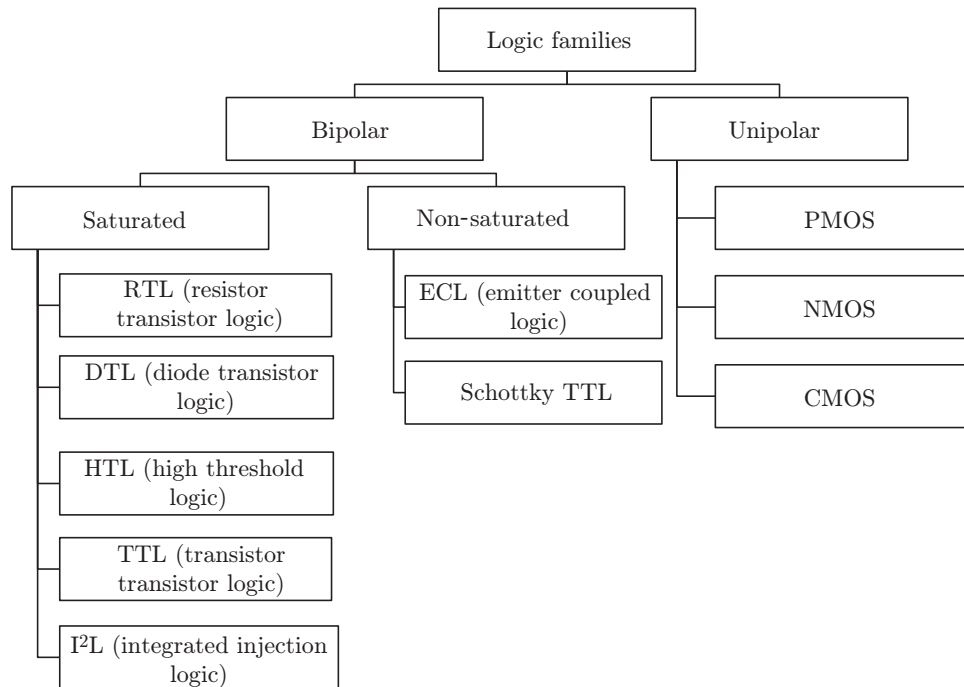
Figure 1.49 | 4-Bit Johnson counter.

Table 1.26 | Truth table of 4-bit Johnson counter

CLK	Q_0	Q_1	Q_2	Q_3	Q'_3
0	0	0	0	0	1
1	1	0	0	0	1
2	1	1	0	0	1
3	1	1	1	0	1
4	1	1	1	1	0
5	0	1	1	1	0
6	0	0	1	1	0
7	0	0	0	1	0
8	0	0	0	0	1

1.5 DIGITAL LOGIC FAMILIES

Refer Fig. 1.50 and Table 1.27 for classification and pros and cons of logic families.

**Figure 1.50** | Digital logic families.**Table 1.27** | Pros and cons of logic families

Logic Families	Pros	Cons
RTL	Economical and simple to design, can easily interface with other logic families	Low threshold and fan out, poor noise immunity, low speed and low power dissipation
DTL	High speed with 30-ns propagation delay, low power dissipation	Poor noise immunity and fan out
TTL	Fastest saturating logic family, high-speed Schottky, low-power Schottky, noise margin about 0.4 V, more fan out, can drive up to 10 gates	Power dissipation is more than MOS gates, noise immunity is not very high
ECL	Highest speed because transistors operate in active region, propagation delay of 2-ns, parameters resistant to temperature change	Very low noise margin, highest power dissipation, high cost
NMOS	Very low power dissipation, large fan out up to 20, very high noise margin	High power dissipation than CMOS, low speed of operation, large propagation delay per gate
I²L	Composed of BJT so high speed of operation, low power dissipation, low power supply requirement	Lower packing density than NMOS, lower noise margin, external resistance required for proper functioning
CMOS	Large fan out (>50), lowest power dissipation, very high noise immunity, temperature resistant	Increased cost, less packing density than NMOS

IMPORTANT FORMULAS

1. Null – All the outputs are zero.
2. Identity – All the outputs are one.
3. X inhibit Y – Represented as $(x \cdot \sim y)$
4. Y inhibit X – Represented as $(y \cdot \sim x)$
5. Transfer – Same as y
6. AND – $x \cdot y$ (Similar to intersection in set theory)
7. OR – $x + y$ (Similar to union in set theory)
8. NOT X – $\sim x$ (Also known as complement of x)
9. NOT Y – $\sim y$ (Also known as complement of y)
10. XOR – $(\sim x \cdot y) + (x \cdot \sim y)$ (Also known as parity checker)
11. XNOR – $(x \cdot y) + (\sim x \cdot \sim y)$
12. NAND – $\sim(x \cdot y)$
13. NOR – $\sim(x + y)$
14. X implication Y ($\sim x + y$)
15. Y implication X ($\sim y + x$)
16. $X + 0 = X$
17. $X + (\sim X) = 1$
18. $X + X = X$
19. $X + 1 = 1$
20. $X + Y = Y + X$
21. $X + (Y + Z) = (X + Y) + Z$
22. $X \cdot (Y + Z) = X \cdot Y + X \cdot Z$
23. $\sim(X + Y) = \sim X \cdot \sim Y$ (Also called **DeMorgan's law**)
24. $X + (X \cdot Y) = X$
25. $\sim(\sim X) = X$

Duality Theorem:

26. $X \cdot 1 = X$
27. $(X \cdot \sim X) = X \cdot X' = 0$

28. $X \cdot X = X$
29. $X \cdot 0 = 0$
30. $X \cdot Y = Y \cdot X$
31. $X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$
32. $X + (Y \cdot Z) = (X + Y) \cdot (X + Z)$
33. $(X \cdot Y)' = X' + Y'$
34. $X \cdot (X + Y) = X$

Half Adder:

35. $S = X'Y + XY'$
36. $C = XY$

Full Adder:

37. $S = X'Y'C_{in} + X'YC'_{in} + XY'C'_{in} + XYC_{in}$
38. $C_{out} = AC_{in} + XY + YC_{in}$

Half Subtractor:

39. $D = XY' + X'Y$
40. $B = X'Y$

Full Subtractor:

41. $D = X'Y'B_{in} + X'YB'_{in} + XY'B'_{in} + XYB_{in}$
42. $B = X'Y + X'YB_{in} + YB_{in}$

Characteristic Equations:

43. SR flip-flop: $Q_{n+1} = S + R'Q_n$
44. D flip-flop: $Q_{n+1} = D$
45. JK flip-flop: $Q_{n+1} = JQ'_n + KQ_n$
46. T flip-flop: $Q_{n+1} = TQ'_n + T'Q_n$

SOLVED EXAMPLES

1. A correct output is achieved from a master–slave JK flip–flop only if its inputs are stable while the
 - (a) Clock is LOW
 - (b) slave is ready to receive the input from master
 - (c) Master gives a reset signal to slave
 - (d) Clock is HIGH

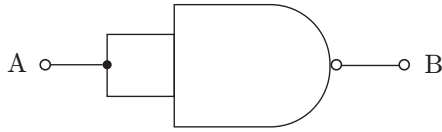
Solution: The clock is high, only the flip–flops are stable when the inputs are stable.

Ans. (d)

2. An inverter circuit can be realized with how many NAND gates?

- (a) 1 (b) 2
(c) 3 (d) Not possible

Solution: An inverter or logic NOT gate can be made using standard NAND gate by connecting together **ALL** their inputs to a common input signal, for example,



Two-input NAND gate equivalent

Ans. (a)

3. A modulus-12 ring counter requires a minimum of _____.

- (a) 10 flip-flops (b) 12 flip-flops
(c) 8 flip-flops (d) 6 flip-flops

Solution: In the ring counter, n flip-flops are used for the mod- n . 12 flip-flops because ring counter uses states n , so here modulus is 12.

Ans. (b)

4. How is a strobe signal used when serially loading a shift register?

- (a) To turn the register set and reset
(b) To control the number of clocks
(c) To determine which output has a HIGH value
(d) To cascade it with other shift register

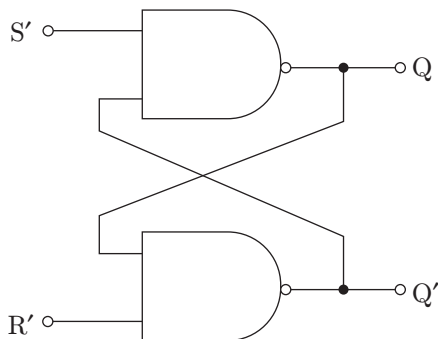
Solution: In digital circuits, a **shift register** is a cascade of flip-flops, sharing the same clock each time. This is called clocking or **strobing** to the register to set the clocks.

Ans. (b)

5. An active-HIGH *SR* latch has a 0 on the *S* input and a 1 on the *R* input. Now when the *R* input goes to 0, the latch will be _____.

- (a) SET (b) RESET
(c) CLR (d) STROBE

Solution:



The *SR* (Set-Reset) flip-flop is a simple sequential circuit which consists of two gates connected as shown above. The output of each gate is connected to one of the inputs of the other gate, thus giving positive feedback or cross-coupling. The circuit has two active low inputs NOT *S* and NOT *R* and two outputs *Q* and NOT *Q*. When there is 0 on the *S* input and a 1 on the *R* input, then NOT *S* is 0 and NOT *R* is 1, so *Q* is 0 and NOT *Q* is 1. Thus, *Q* is reset to 0 by 0 on NOT *R*.

Ans. (b)

6. The NAND or NOR gates are referred to as “universal” gates because

- (a) they are easier to implement.
(b) they can be used to build all the other types of gates.
(c) they are available in different IC families.
(d) they are available everywhere in the world.

Solution: NAND and NOR gates are universal because all Boolean functions that can be implemented by AND, OR and NOT gates can be implemented by NAND and NOR gates.

Ans. (b)

7. Match the following:

IC Family	GATE
TTL	NOR
CMOS	NOT
ECL	NAND

- (a) TTL:NAND; ECL:NOR; CMOS:NOT
(b) TTL:NAND; ECL:NOT; CMOS:NOR
(c) TTL:NOT; ECL:NAND; CMOS:NOR
(d) TTL:NOR; ECL:NOT; CMOS:NAND

Solution: The NAND function is actually the simplest, most natural mode of operation for this TTL design. The basic gates are OR/NOR gates for ECL Construction. Inverters can be constructed using two complementary transistors in a CMOS configuration.

Ans. (a)

8. Which of the following satisfies the commutative law but not the associative law?

- (a) NOR (b) OR
(c) XOR (d) NAND

Solution

Let $\uparrow$ signify the NAND operation. Then there exist propositions p, q, r such that

$$p \uparrow (q \uparrow r) \neq (p \uparrow q) \uparrow r$$

Solution:

x	y	z	$x \oplus y$	$(x \oplus y) \oplus z$	$y \oplus z$	$x \oplus (y \oplus z)$
0	0	0	1	0	1	0
0	0	1	1	1	0	1
0	1	0	0	1	0	1
0	1	1	0	0	1	0
1	0	0	0	1	1	1
1	0	1	0	0	0	0
1	1	0	1	0	0	0
1	1	1	1	1	1	1

Ans. (b)

17. Let $*$ be defined as $x * y = \bar{x} + y$. Let $z = x * y$, then value of $z * x$ is

(a) $\bar{x} + y$ (b) x (c) 0 (d) 1

Solution: Substituting value of $z = x * y$ in the expression:

$$\begin{aligned} z * x &= (x * y) * x = (\bar{x} + y) * x = \overline{(\bar{x} + y)} + x \\ &= x \cdot \bar{y} + x \end{aligned}$$

Now using absorption law: $a + ab = a$

$$x \cdot \bar{y} + x = x$$

So, result is: $z * x = x$

Ans. (b)

18. Let $f(x, y, z) = \bar{x} + \bar{y}z + xz$ be a switching function. Which one of the following is valid?

(a) $\bar{y}x$ is a prime implicant of f .
 (b) xz is a minterm of f .
 (c) xz is an implicant of f .
 (d) y is a prime implicant of f .

Solution:

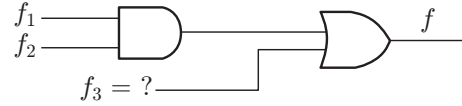
x \ yz	00	01	11	10
0	1	1	1	1
1		1	1	

Number of possible joins are 2, so there are two prime implicants, that is, x' and z .

Both the prime implicants are essential. xz is an implicant of f .

Ans. (c)

19. Consider the logic circuit shown in the following figure. The functions f_1, f_2 and f (in canonical sum-of-products form in decimal notation) are



$$f_1(w, x, y, z) = \sum (8, 9, 10)$$

$$f_2(w, x, y, z) = \sum (7, 8, 12, 13, 14, 15)$$

$$f(w, x, y, z) = \sum (8, 9)$$

The function f_3 is

(a) $\sum 9, 10$

(b) $\sum 9$

(c) $\sum 1, 8, 9$

(d) $\sum 8, 10, 18$

Solution: By the logic circuit, function f is given as

$$f = f_1 f_2 + f_3$$

$$f_1 = \sum (8, 9, 10)$$

$$f_2 = \sum (7, 8, 12, 13, 14, 15)$$

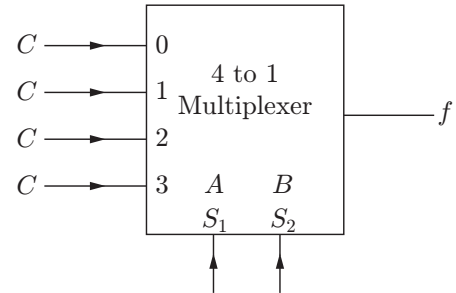
As according to logic, common minterm in f_1 and f_2 is 8, so

$$f_1 f_2 = \sum (8)$$

to get output as $f = \sum (8, 9)$, f_3 can be either $\sum (8, 9)$ or $f_3 = \sum 9$

Ans. (b)

20. Consider the circuit shown in the following figure; f implements



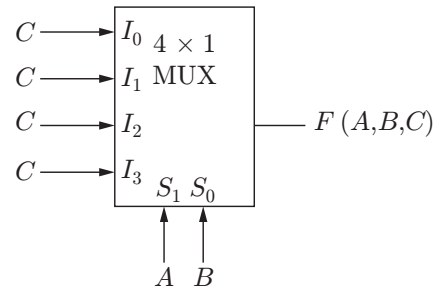
(a) $\bar{A}\bar{B}C + \bar{A}B\bar{C} + ABC$

(b) $A + B + C$

(c) $A \oplus B \oplus C$

(d) C

Solution: The circuit for function $f(A, B, C)$ is given as follows



The output of function will be

$$\begin{aligned} \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + ABC &= \bar{A}C(\bar{B} + B) + AC(\bar{B} + B) \\ &= \bar{A}C + AC = C \end{aligned}$$

Ans. (d)

21. Which of the following functions implements the Karnaugh map shown below?

AB \ CD	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

- (a) $\bar{A}B + CD$ (b) $D(C + A)$
(c) $AD + \bar{A}B$ (d) $(C + D)(\bar{C} + D)(A + B)$

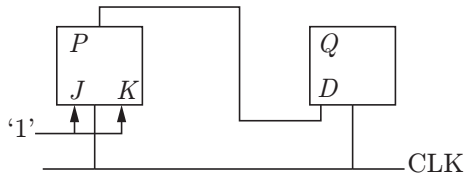
Solution: Karnaugh map of the function is given as follows:

AB \ CD	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

SOP form for the given function is $f(A, B, C, D) = CD + AD = D(C + A)$

Ans. (b)

22. Refer to the following arrangement of master–slave flip–flops.



It has the initial state of P, Q as 0, 1 (respectively). After the clock cycles, the output state P, Q is (respectively)

- (a) 1, 0 (b) 1, 1 (c) 0, 0 (d) 0, 1

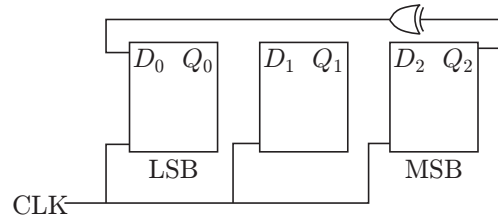
Solution:

	J	K	D	P	Q
	1	1	(P)	P	Q
Initially				0	1
	1	1	0	1	0
	1	1	1	0	1

For $J = 1$ and $K = 1$, the output is complement to previous state.

Ans. (a)

23. Consider the circuit given below with initial state $Q_0 = 1, Q_1 = Q_2 = 0$. The state of the circuit is given by the value $4Q_2 + 2Q_1 + Q_0$.



Which one of the following is the correct state sequence of the circuit?

- (a) 1, 3, 4, 6, 7, 5, 2 (b) 1, 2, 5, 3, 7, 6, 4
(c) 1, 2, 7, 3, 5, 6, 4 (d) 1, 6, 5, 7, 2, 3, 4

Solution:

$D_2(Q_1)$	$D_1(Q_0)$	D_0 $(Q_1 \oplus Q_0)$	Q_2	Q_1	Q_0	$4Q_2 + 2Q_1 + Q_0$
Initially			0	0	1	1
0	1	0	0	1	0	2
1	0	1	1	0	1	5
0	1	1	0	1	1	3
1	1	1	1	1	1	7
1	1	0	1	1	0	6
1	0	0	1	0	0	4
0	0	1	0	0	1	1

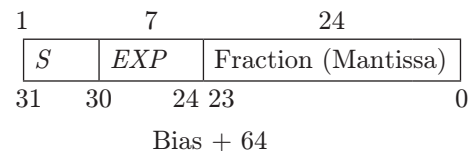
Therefore, the correct state sequence of the circuit is 1, 2, 5, 3, 7, 6, 4.

Ans. (b)

24. A 32-bit floating-point number is represented by a 7-bit signed exponent, and a 24-bit fractional mantissa. The base of the scale factor is 16.

- (a) The range of the exponent is _____.
(b) The range of the exponent is _____ if the scale factor is represented in excess –64 format.

Solution: The given floating point number format is



The above floating-point architecture is IBM floating-point architecture.

The number is represented as the following formula
 $(-1)^S \times 0. \text{ Mantissa} \times 10^{\text{EXP} - 64}$

- (a) The range of the exponent is the range of the 7-bit signed numbers, which is –64 to +63.
(b) The range of the exponent if the scale factor is represented in excess –64 format is 0 to 127.

25. Booth's coding in 8-bits for the decimal number -57 is

- (a) $0 - 100 + 1000$ (b) $0 - 100 + 100 - 1$
 (c) $0 - 1 + 100 - 10 + 1$ (d) $00 - 10 + 100 - 1$

Solution: -57 can be represented in 2's complement form: 1 1 0 0 0 1 1 1

Booth's coding for decimal number -57 is

$$\begin{array}{r} 1 \quad 1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ \boxed{0} \\ \downarrow \\ 0 \quad -1 \ 0 \ 0 \ +1 \ 0 \ 0 \ -1 \ -1 \end{array}$$

Ans. (b)

26. The 2' complement representation of the decimal value -15 is

- (a) 1111 (b) 11111
 (c) 111111 (d) 10001

Solution: $+15 = 1 \ 1 \ 1 \ 1$
 -15 in 2's complement = 10001

Ans. (d)

GATE PREVIOUS YEARS' QUESTIONS

1. Assuming all numbers are in 2's complement representation, which of the following numbers is divisible by 11111011?

- (a) 11100111 (b) 11100100
 (c) 11010111 (d) 11011011

(GATE 2003: 1 Mark)

Solution:

2's Complement	Binary Number	Decimal Value
11100111	00011001	25
11100100	00011100	28
11010111	00101001	41
11011011	00100101	37

11111011 is 5 in decimal. Only 25 is divisible by 5.

Ans. (a)

2. The following is a scheme for floating point number representation using 16 bits.

Bit Position	15	14.....9	8.....0
	s	e	m
	sign	exponent	mantissa

Let s , e , and m be the numbers represented in binary in the sign, exponent, and mantissa fields, respectively. Then the floating point number represented is

$$\begin{cases} (-1)^s (1 + m \times 2^{-9}) 2^{e-31} & \text{if the exponent} \neq 11111 \\ 0 & \text{otherwise} \end{cases}$$

What is the maximum difference between two successive real numbers representable in this system?

- (a) 2^{-40} (b) 2^{-9}
 (c) 2^{22} (d) 2^{31}

(GATE 2003: 2 Marks)

Solution:

Largest positive number, $m = 11111111$

exponent, $e = 111110$

Second largest positive number = 11111110

exponent, $e = 111110$

Difference = $2^{31} (2 - 2^{-9} - 2 + 2^{-8}) = 2^{22}$

Ans. (c)

3. Consider an array multiplier for multiplying two n bit numbers. If each gate in the circuit has a unit delay, the total delay of the multiplier is

- (a) $\Theta(1)$ (b) $\Theta(\log n)$ (c) $\Theta(n)$ (d) $\Theta(n^2)$

(GATE 2003: 1 Mark)

Solution:

$$\text{Delay} = 1 + 2 + \dots + n = \frac{n(n+1)}{2} = \Theta(n^2)$$

Ans. (d)

4. A 1-input, 2-output synchronous sequential circuit behaves as follows:

Let z_k , n_k denote the number of 0's and 1's, respectively, in initial k bits of the input ($z_k + n_k = k$). The circuit outputs 00 until one of the following conditions holds.

- (a) $z_k - n_k = 2$. In this case, the output at the k th and all subsequent clock ticks is 10.
 (b) $n_k - z_k = 2$. In this case, the output at the k th and all subsequent clock ticks is 01.

What is the minimum number of states required in the state transition graph of the above circuit?

- (a) 5 (b) 6 (c) 7 (d) 8

(GATE 2003: 2 Marks)

Solution: Number of transition states are 7.

Ans. (c)

5. The literal count of a Boolean expression is the sum of the number of times each literal appears in the expression. For example, the literal count of $(xy + xz')$ is 4. What are the minimum possible literal counts of the product-of-sum and sum-of-product representations, respectively, of the function given by the following Karnaugh map? Here, X denotes “don’t care”.

ZY XY	ZW			
	00	01	11	10
00	X	1	0	1
01	0	1	X	0
11	1	X	X	0
10	X	0	0	X

- (a) (11, 9) (b) (9, 13)
(c) (9, 10) (d) (11, 11)

(GATE 2003: 2 Marks)

Solution:

ZY XY	ZW			
	00	01	11	10
00	X	1	0	1
01	0	1	X	0
11	1	X	X	0
10	X	0	0	X

SOP: $wy + w'y' + z'wx' + xyz'$

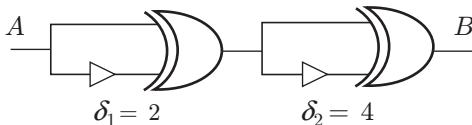
Literal count: 10

POS: $(y' + z')(w' + z')(z' + y)(x + z + w)$

Literal count: 9

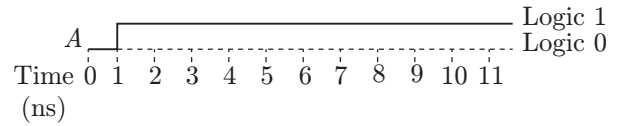
Ans. (c)

6. Consider the following circuit composed of XOR gates and non-inverting buffers.



The non-inverting buffers have delays $\delta_1 = 2$ ns and $\delta_2 = 4$ ns as shown in the figure. Both XOR gates and all wires have zero delay. Assume that all gate inputs, outputs and wires are stable at logic level 0 at time 0. If the following waveform is applied at

input A, how many transition(s) (change of logic levels) occur(s) at B during the interval from 0 to 10 ns?



- (a) 1 (b) 2
(c) 3 (d) 4

(GATE 2003: 2 Marks)

Solution: Total number of transitions is 3.

1: 0 to 0

2: 0 to 1

3: 1 to 1

Ans. (c)

7. A Boolean function $x'y' + xy + x'y$ is equivalent to

- (a) $x' + y'$ (b) $x + y$
(c) $x + y'$ (d) $x' + y$

(GATE 2004: 1 Mark)

Solution:

$x'y' + xy + x'y$

$= x'(y' + y) + xy$ (as $y' + y = 1$)

$= x' + xy$ (as $x + x'y = x + y$)

$= x' + y$

Ans. (d)

8. In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in

- (a) $Q = 0, Q' = 1$ (b) $Q = 1, Q' = 0$
(c) $Q = 1, Q' = 1$ (d) Indeterminate states

(GATE 2004: 1 Mark)

Solution: If both S and R are 0, using SR latch with NAND gate output state will be invalid state.

Ans. (d)

9. If 73_x (in base-x number system) is equal to 54_y (in base-y number system), the possible values of x and y are

- (a) 8, 16 (b) 10, 12
(c) 9, 13 (d) 8, 11

(GATE 2004: 1 Mark)

Solution: By doing hit and trial with the options, we arrive at

If $x = 8$, $(73)_8 = 7 \times 8^1 + 3 \times 8^0 = 59$

If $y = 11$, $(54)_{11} = 5 \times 11^1 + 4 \times 11^0 = 59$

Ans. (d)

10. What is the result of evaluating the following two expressions using three-digit floating point arithmetic with rounding?

$$(113 + -111) + 7.51$$

$$113 + (-111 + 7.51)$$

- (a) 9.51 and 10.0, respectively
 (b) 10.0 and 9.51, respectively
 (c) 9.51 and 9.51, respectively
 (d) 10.0 and 10.0, respectively

(GATE 2004: 1 Mark)

Solution:

$$(113 + -111) + 7.51 = 9.51$$

$$113 + (-111 + 7.51) = 10.0$$

Ans. (a)

11. Consider a multiplexer with X and Y as data inputs and Z as control input. $Z = 0$ selects input X, and $Z = 1$ selects input Y. What are the connections required to realize the 2-variable Boolean function $f = T + R$, without using any additional hardware?

- (a) R to X, 1 to Y, T to Z
 (b) T to X, R to Y, T to Z
 (c) T to X, R to Y, 0 to Z
 (d) R to X, 0 to Y, T to Z

(GATE 2004: 2 Marks)

Solution: Connect R to X, 1 to Y and T to Z.

Ans. (a)

12. A circuit outputs a digit in the form of 4 bits, where 0 is represented by 0000, 1 by 0001, ..., 9 by 1001. A combinational circuit is to be designed which takes these 4 bits as input and output 1 if the digit ≥ 5 , and 0 otherwise. If only AND, OR and NOT gates are used, what is the minimum number of gates required?

- (a) 2
 (b) 3
 (c) 4
 (d) 5

(GATE 2004: 2 Marks)

Solution:

a	B	c	d	Output
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	0	1	0
0	1	0	0	0
0	1	1	1	1

(Continued)

Continued

A	B	c	d	Output
0	1	1	0	1
0	1	0	1	1
1	0	0	0	1
1	0	1	1	1

Now using K-map, the expression becomes $a + bd + bc \rightarrow a + b(d+c)$.

Hence, three gates will be required.

Ans. (b)

13. Which are the essential prime implicants of the following Boolean function?

$$f(a, b, c) = a'c + ac' + b'c$$

- (a) $a'c$ and ac'
 (b) $a'c$ and $b'c$
 (c) $a'c$ only
 (d) ac' and bc'

(GATE 2004: 2 Marks)

Solution:

$$f(a, b, c) = a'c + ac' + b'c$$

$$= a'(b + b')c + a(b + b')c' + (a + a')b'c$$

$$= a'bc + a'b'c + abc' + ab'c' + a'b'c$$

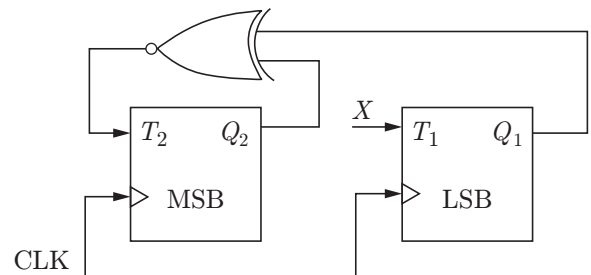
Using K-map:

c \ ab	00	01	11	10
0			1	1
1	1	1		

here ca' and $c'a$ are essential prime implicants, we cannot remove them.

Ans. (a)

14. Consider the partial implementation of a 2-bit counter using T flip-flops following the sequence 0-2-3-1-0, as shown in the following figure.



To complete the circuit, the input X should be

- (a) Q_2'
 (b) $Q_2 + Q_1$
 (c) $(Q_1 \oplus Q_2)'$
 (d) $Q_1 \oplus Q_2$

(GATE 2004: 2 Marks)

Solution: Sequence 0 – 2 – 3 – 1 – 0 means 00 – 11 – 01 – 00

If $x = Q_1 \oplus Q_2$, then the given output will be produced.

Ans. (d)

15. A 4-bit carry look-ahead adder, which adds two 4-bit numbers, is designed using AND, OR, NOT, NAND, NOR gates only. Assuming that all the inputs are available in both complemented and uncomplemented forms and the delay of each gate is one time unit, what is the overall propagation delay of the adder? Assume that the carry network has been implemented using two-level AND–OR logic.

- (a) 4 time units (b) 6 time units
(c) 10 time units (d) 12 time units

(GATE 2004: 2 Marks)

Solution: Equations for carry look-ahead adder are as follows:

$$C_1 = G_0 + P_0 \cdot C_0$$

$$C_2 = G_1 + P_1 \cdot C_1$$

$$C_3 = G_2 + P_2 \cdot C_2$$

$$C_4 = G_3 + P_3 \cdot C_3$$

The total number of gate delay will be 6.

Ans. (b)

16. Let $A = 1111\ 1010$ and $B = 0000\ 1010$ be two 8-bit 2's complement numbers. Their product in 2's complement is

- (a) 1100 0100 (b) 1001 1100
(c) 1010 0101 (d) 1101 0101

(GATE 2004: 2 Marks)

Solution:

$$A = 1111\ 1010 = 0000\ 0110 = -6$$

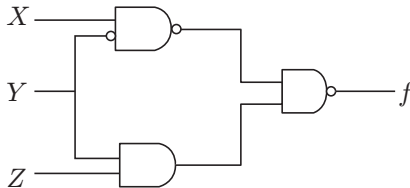
$$B = 0000\ 1010 = 10$$

$$A \times B = -60$$

$$-60 = 11000100$$

Ans. (a)

17. Consider the following circuit.



Which one of the following is TRUE?

- (a) f is independent of X .
(b) f is independent of Y .

(c) f is independent of Z .

(d) None of X , Y , Z is redundant.

(GATE 2005: 1 Mark)

Solution:

$$\begin{aligned} f &= XY' + (YZ)' \\ &= Y' + Z' \end{aligned}$$

So, f is independent of X .

Ans. (a)

18. The range of integers that can be represented by an n -bit 2's complement number system is

- (a) -2^{n-1} to $(2^{n-1}-1)$
(b) $-2(2^{n-1}-1)$ to $(2^{n-1}-1)$
(c) -2^{n-1} to 2^{n-1}
(d) $-2(2^{n-1}+1)$ to $(2^{n-1}-1)$

(GATE 2005: 1 Mark)

Solution: Negative numbers are one more than positive number. So, the range of 2's complement is 2^{n-1} to $(2^{n-1}-1)$.

Ans. (a)

19. The hexadecimal representation of 657_8 is

- (a) 1AF (b) D78
(c) D71 (d) 32F

(GATE 2005: 1 Mark)

Solution:

$$(657)_8 = (110\ 101\ 111)_8$$

$$(0001\ 1010\ 1111)_{16} = (1AF)_{16}$$

Ans. (a)

20. The switching expression corresponding to $f(A, B, C, D) = \sum(1, 4, 5, 9, 11, 12)$ is

- (a) $BC'D' + A'C'D + AB'D$
(b) $ABC' + ACD + B'C'D$
(c) $ACD' + A'BC' + AC'D'$
(d) $A'BD + ACD' + BCD'$

(GATE 2005: 1 Mark)

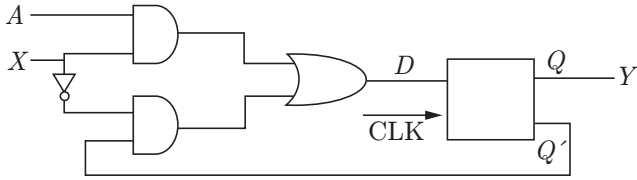
Solution:

$AB \backslash CD$	00	01	11	10
00		1		
01	1	1		
11	1			
10		1	1	

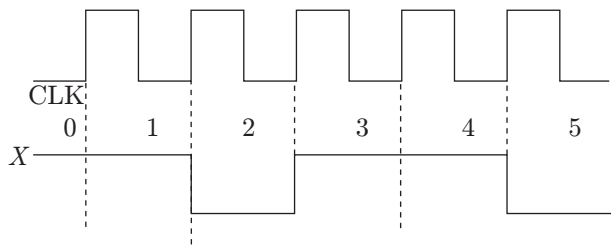
$$f = BC'D' + A'C'D + AB'D$$

Ans. (a)

21. Consider the following circuit involving a positive-edge triggered D FF.



Consider the following timing diagram. Let A_i represent the logic level on the line A in the i th clock period.



Let A' represent the complement of A . The correct output sequence on Y over the clock periods 1 through 5 is

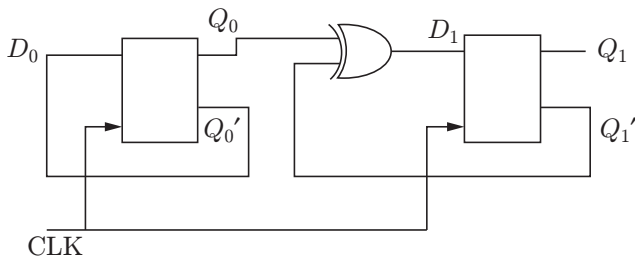
- (a) $A_0A_1A_1'A_3A_4$ (b) $A_0A_1A_2'A_3A_4$
(c) $A_1A_2A_2'A_3A_4$ (d) $A_1A_2'A_3A_4A_5'$

(GATE 2005: 2 Marks)

Solution: The correct output sequence on Y is $A_0A_1A_1'A_3A_4$.

Ans. (a)

22. Consider the following circuit.



The flip-flops are positive-edge triggered D FFs. Each state is designated as a 2-bit string Q_0Q_1 . Let the initial state be 00. The state transition sequence is

- (a) 00-11-01 (b) 00-11
(c) 00-10-01-11 (d) 00-11-01-10

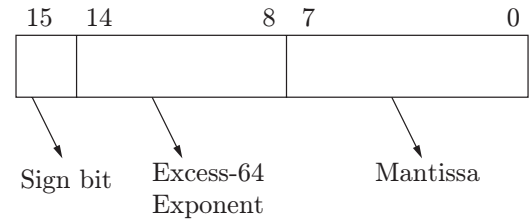
(GATE 2005: 2 Marks)

Solution:

D_0	D_1	Q_0	Q_1
1	1	1	1
0	1	0	1
1	0	1	0
0	0	0	0

Ans. (d)

Linked Answer Questions 23 and 24: Consider the following floating-point format.



Mantissa is a pure fraction in sign-magnitude form.

23. The decimal number 0.239×2^{13} has the following hexadecimal representation (without normalization and rounding off):

- (a) 0D 24 (b) 0D 4D
(c) 4D 0D (d) 4D 3D

(GATE 2005: 2 Marks)

Solution:

$$0.239 \times 2^{13} = 0.00111101 \times 2^{13}$$

$$64 + 13 = 77$$

0	100 1101	0011 1101
	4D	3D

Ans. (d)

24. The normalized representation for the above format is specified as follows. The mantissa has an implicit 1 preceding the binary (radix) point. Assume that only 0's are padded in while shifting a field. The normalized representation of the above number (0.239×2^{13}) is

- (a) 0A 20 (b) 11 34
(c) 49 D0 (d) 4A E8

(GATE 2005: 2 Marks)

Solution: Using normalized form

$$1.11101 \times 2^{10}$$

$$64 + 10 = 74$$

0	100 1010	11101000
	4A	E8

Ans. (d)

Solution:

Clock	Input	q_2	q_1	q_0
0	1	0	0	1
1	0	0	1	0
2	0	1	0	0
3	1	0	1	1
4	1	1	1	1
5	0	1	0	0
6	0	0	1	0
7	0	1	0	0
8	0	0	1	0

Ans. (c)

29. Consider a Boolean function $f(w, x, y, z)$. Suppose that exactly one of its inputs is allowed to change at a time. If the function happens to be true for two input vectors $i_1 = (w_1, x_1, y_1, z_1)$ and $i_2 = (w_2, x_2, y_2, z_2)$, we would like the function to remain true as the input changes from i_1 to i_2 (i_1 and i_2 differ in exactly 1 bit position), without becoming false momentarily. Let $f(w, x, y, z) = (5, 7, 11, 12, 13, 15)$. Which of the following cube covers of f will ensure that the required property is satisfied?

- (a) $w'xz, wxy', xy'z, xyz, wyz$
 (b) $wxy, w'xz, wyz$
 (c) $wxyz, xz, wx'yz$
 (d) $wzy, wyz, wxz, w'xz, xy'z, xyz$

(GATE 2006: 1 Mark)

Solution:

$yz \backslash wx$	00	01	11	10
00				
01		1	1	
11	1	1	1	
10			1	

From the K-map, we have $w'xz, wxy', xy'z, xyz, wyz$.

Ans. (a)

30. We consider the addition of two 2's complement numbers $b_{n-1}b_{n-2} \dots b_0$ and $a_{n-1}a_{n-2} \dots a_0$. A binary adder for adding unsigned binary numbers is used to add the two numbers. The sum is denoted by $c_{n-1}c_{n-2} \dots c_0$ and the carry-out by c_{out} .

Which one of the following options correctly identifies the overflow condition?

- (a) $\overline{c_{out}(a_{n-1} \oplus b_{n-1})}$
 (b) $\overline{a_{n-1}b_{n-1}c_{n-1}} + \overline{a_{n-1}b_{n-1}c_{n-1}}$

(c) $c_{out} \oplus c_{n-1}$

(d) $a_{n-1} \oplus b_{n-1} \oplus c_{n-1}$

(GATE 2006: 1 Mark)

Solution: The overflow will be identified c_{out} and c_{n-1} are XORed.

Ans. (c)

31. Consider numbers represented in 4-bit gray code. Let $h_3h_2h_1h_0$ be the gray code representation of a number n and let $g_3g_2g_1g_0$ be the gray code of $(n+1)$ (modulo 16) value of the number. Which one of the following functions is correct?

(a) $g_0(h_3h_2h_1h_0) = \sum(1, 2, 3, 6, 10, 13, 14, 15)$

(b) $g_1(h_3h_2h_1h_0) = \sum(4, 9, 10, 11, 12, 13, 14, 15)$

(c) $g_2(h_3h_2h_1h_0) = \sum(2, 4, 5, 6, 7, 12, 13, 15)$

(d) $g_3(h_3h_2h_1h_0) = \sum(0, 1, 6, 7, 10, 11, 12, 13)$

(GATE 2006: 1 Mark)

Solution: $(n+1) \bmod 16$ is a 5-bit number; range is from 0 to 31.

$$g_2(2,4,5,6,7,12,13,15)$$

Ans. (c)

32. What is the maximum number of different Boolean functions involving n Boolean variables?

(a) n^2

(b) 2^n

(c) 2^{2^n}

(d) 2^{n^2}

(GATE 2007: 1 Mark)

Solution: Maximum Boolean functions with n elements can be 2^{2^n} .

Ans. (c)

33. How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

(a) 7

(b) 8

(c) 9

(d) 10

(GATE 2007: 1 Mark)

Solution: We need to construct 6 to 64 decoder so apply the following:

$$\sum_{i=1}^{\log 64} \frac{64}{8^i} = \frac{64}{8} + \frac{64}{64} = 9$$

Ans. (c)

34. Consider the following Boolean function of four variables:

$$f(w, x, y, z) = \sum(1, 3, 4, 6, 9, 11, 12, 14)$$

The function is

(a) independent of one variable

(b) independent of two variables

(c) independent of three variables

(d) dependent on all the variables

(GATE 2007: 1 Mark)

Solution: $f(w, x, y, z) = \sum(1, 3, 4, 6, 9, 11, 12, 14)$ can be represented in K-map as

$wx \backslash yz$	00	01	11	10
00		1	1	
01	1			1
11	1			1
10		1	1	

Quadruple is formed, indicating the function is independent of two variables.

Ans. (b)

35. Let $f(w, x, y, z) = \sum(0, 4, 5, 7, 8, 9, 13, 15)$. Which of the following expressions are NOT equivalent to f ?

- (P) $x'y'z' + w'xy' + wy'z + xz$
 (Q) $w'y'z' + wx'y' + xz$
 (R) $w'y'z' + wx'y' + xyz + xy'z$
 (S) $x'y'z' + wx'y' + w'y$

- (a) P only (b) Q and S
 (c) R and S (d) S only

(GATE 2007: 2 Marks)

Solution:

$wx \backslash yz$	00	01	11	10
00	1			
01	1	1	1	
11		1	1	
10	1	1		

Option (d) does not contain essential prime implicants.

Ans. (d)

36. Define the connective $*$ for the Boolean variables X and Y as $X*Y = XY + X'Y'$. Let $Z = X*Y$. Consider the following expressions P , Q and R .

$$P: X = Y*Z \quad Q: Y = X*Z \quad R: X*Y*Z = 1$$

Which of the following is TRUE?

- (a) Only P and Q are valid.
 (b) Only Q and R are valid.
 (c) Only P and R are valid.
 (d) All P, Q, R are valid.

(GATE 2007: 2 Marks)

Solution:

$$P: Y*Z = YZ + Y'Z' = Y(XY + X'Y') + Y'(X'Y + XY') = X$$

So, P is valid.

$$Q: Y = X*Z$$

$$X*Z = XZ + X'Z' = X(XY + X'Y') + X'(X'Y + XY') = Y$$

So, Q is valid.

$$R: X*Y*Z = 1$$

$$X*Y*Z = Z*Z = ZZ + Z'Z' = 1$$

So, R is also valid.

Ans. (d)

37. Suppose only one multiplexer and one inverter are allowed to be used to implement any Boolean function of n variables. What is the minimum size of the multiplexer needed?

- (a) 2^n line to 1 line (b) 2^{n+1} line to 1 line
 (c) 2^{n-1} line to 1 line (d) 2^{n-2} line to 1 line

(GATE 2007: 2 Marks)

Solution: Since only one inverter is allowed, so $n-1$ variable will be used for selection and one as input. 2^{n-1} line to 1 line multiplexer will be required.

Ans. (c)

38. In a look-ahead carry generator, the carry generate function G_i and the carry propagate function P_i for inputs A_i and B_i are given by

$$P_i = A_i \oplus B_i \text{ and } G_i = A_i B_i$$

The expressions for the sum bit S_i and the carry bit C_{i+1} of the look-ahead carry adder are given by $S_i = P_i \oplus C_i$ and $C_{i+1} = G_i + P_i C_i$, where C_0 is the input carry.

Consider a two-level logic implementation of the look-ahead carry generator. Assume that all P_i and G_i are available for the carry generator circuit and that the AND and OR gates can have any number of inputs. The number of AND gates and OR gates needed to implement the look-ahead carry generator for a 4-bit adder with S_3 , S_2S_1 , S_0 , and C_4 as its outputs are, respectively,

- (a) 6, 3 (b) 10, 4
 (c) 6, 4 (d) 10, 5

(GATE 2007: 2 Marks)

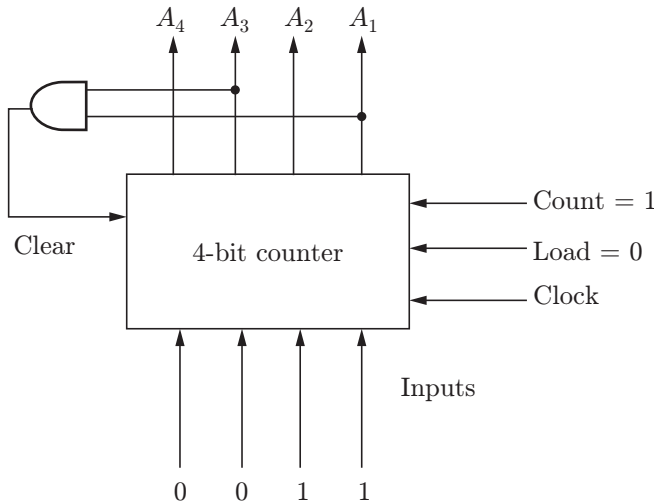
Solution: Two 1-bit adders are required, where each 1-bit adder required 5 AND and 2 OR gates. So, 10 AND gates and 4 OR gates are required.

Ans. (b)

39. The control signal functions of a 4-bit binary counter are given below (where X is "don't care"):

Clear	Clock	Load	Count	Function
1	X	X	X	Clear to 0
0	X	0	0	No change
0		1	X	Load input
0		0	1	Count next

The counter is connected as follows:



Assume that the counter and gate delays are negligible. If the counter starts at 0, then it cycles through the following sequence:

- (a) 0, 3, 4 (b) 0, 3, 4, 5
 (c) 0, 1, 2, 3, 4 (d) 0, 1, 2, 3, 4, 5

(GATE 2007: 2 Marks)

Solution: The counter will count if count = 1 and load = 0, the sequence will be 0, 1, 2, 3, 4.

Ans. (c)

40. In the Karnaugh map shown below, X denotes a don't care term. What is the minimal form of the function represented by the Karnaugh map?

	ab			
cd	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

- (a) $\bar{b} \cdot \bar{d} + \bar{a} \cdot \bar{d}$ (b) $\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{d} + \bar{a} \cdot \bar{d}$
 (c) $\bar{b} \cdot \bar{d} + \bar{a} \cdot b \cdot \bar{d}$ (d) $\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{d} + \bar{a} \cdot \bar{d}$

(GATE 2008: 1 Mark)

Solution:

	ab			
cd	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

$$\text{So, } \bar{b} \cdot \bar{d} + \bar{a} \cdot \bar{d}$$

Ans. (a)

41. In the IEEE floating-point representation, the hexadecimal value 0x00000000 corresponds to

- (a) The normalized value 2^{-127}
 (b) The normalized value 2^{-126}
 (c) The normalized value $+0$
 (d) The special value $+0$

(GATE 2008: 1 Mark)

Solution: IEEE floating point standard represent number in normalized form. This is the smallest positive number, that is, $+0$.

Ans. (d)

42. Let r denote number system radix. The only value(s) of r that satisfy the equation $\sqrt{121_r} = 11_r$ is/are

- (a) Decimal 10 (b) Decimal 11
 (c) Decimal 10 and 11 (d) Any value > 2

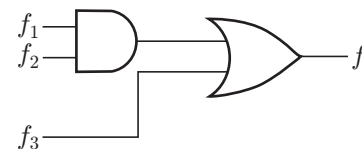
(GATE 2008: 1 Mark)

Solution:

$$\begin{aligned} (1 + 2r^2 + r^3)^{1/2} &= 1 + r \\ \Rightarrow 1 + 2r^2 + r^3 &= (1 + r)^2 = 1 + r^2 + 2r \\ r^3 + r^2 - 2r &= 0 \Rightarrow r > 2 \end{aligned}$$

Ans. (d)

43. Given f_1, f_3 and f in canonical sum of products form (in decimal) for the circuit



$$\begin{aligned} f_1 &= \sum m(4, 5, 6, 7, 8) \\ f_3 &= \sum m(1, 6, 15) \\ f &= \sum m(1, 6, 8, 15) \end{aligned}$$

then f_2 is

- (a) $\sum m(4, 6)$ (b) $\sum m(4, 8)$
 (c) $\sum m(6, 8)$ (d) $\sum m(4, 6, 8)$

(GATE 2008: 1 Mark)

Solution: To get $f = \sum m(1, 6, 8, 15)$, f_2 should set 6 and 8.

Ans. (c)

44. If
- P, Q, R
- are Boolean variables, then

$$(P + \bar{Q})(P \cdot \bar{Q} + P \cdot R)(\bar{P} \cdot \bar{R} + \bar{Q})$$

simplifies to

- (a) $P \cdot \bar{Q}$ (b) $P \cdot \bar{R}$
 (c) $P \cdot \bar{Q} + R$ (d) $P \cdot \bar{R} + Q$

(GATE 2008: 2 Marks)

Solution:

$$\begin{aligned} (P + \bar{Q})(P \cdot \bar{Q} + P \cdot R)(\bar{P} \cdot \bar{R} + \bar{Q}) \\ = (P \cdot \bar{Q} + P \cdot \bar{Q} + P \cdot \bar{Q} \cdot R + PR)(\bar{P} \cdot \bar{R} + \bar{Q}) \\ = P \cdot \bar{Q} + P \cdot \bar{Q} + P \cdot \bar{Q} \cdot R + P \cdot \bar{Q} \cdot R \\ = P \cdot \bar{Q} \quad (\text{as } 1 + R = 1) \end{aligned}$$

Ans. (a)

- 45.
- $(1217)_8$
- is equivalent to

- (a) $(1217)_{16}$ (b) $(028F)_{16}$
 (c) $(2297)_{10}$ (d) $(0B17)_{16}$

(GATE 2009: 1 Mark)

Solution:

$$\begin{aligned} (1217)_8 &= (001\ 010\ 001\ 111)_8 \\ (0010\ 1000\ 1111) &= (28F)_{16} = (028F)_{16} \end{aligned}$$

Ans. (b)

46. What is the minimum number of gates required to implement the Boolean function
- $(AB + C)$
- if we have to use only 2-input NOR gates?

- (a) 2 (b) 3
 (c) 4 (d) 5

(GATE 2009: 1 Mark)

Solution:

$$\begin{aligned} (AB + C) &= (A + C)(B + C) \\ &= ((A + C)' + (B + C)')' \end{aligned}$$

Thus, it requires three NOR gates.

Ans. (b)

47. The binary operation
- $\square$
- is defined as follows

P	Q	$P \square Q$
T	T	T
T	F	T
F	T	F
F	F	T

Which one of the following is equivalent to $P \vee Q$?

- (a) $\neg Q \square \neg P$ (b) $P \square \neg Q$
 (c) $\neg P \square Q$ (d) $\neg P \square \neg Q$

(GATE 2009: 1 Mark)

Solution: Try using different values of P and Q .
 For example, $P = \text{True}$ and $Q = \text{False}$.

Ans. (b)

48. How many
- $32\text{ K} \times 1$
- RAM chips are needed to provide a memory capacity of 256 K-bytes?

- (a) 8 (b) 32
 (c) 64 (d) 128

(GATE 2009: 1 Mark)

Solution:

$$\text{Number of chips required} = \frac{256 \times 8}{32 \times 1} = 64$$

Ans. (c)

49. The minterm expansion of
- $f(P, Q, R) = PQ + Q\bar{R} + P\bar{R}$
- is

- (a) $m_2 + m_4 + m_6 + m_7$ (b) $m_0 + m_1 + m_3 + m_5$
 (c) $m_0 + m_1 + m_6 + m_7$ (d) $m_2 + m_3 + m_4 + m_5$

(GATE 2010: 1 Mark)

Solution:

$$\begin{aligned} f(P, Q, R) &= PQ + QR' + PR' \\ &= PQ(R + R') + (P + P')QR' + P(Q + Q')R' \\ &= PQR + PQR' + P'QR' + PQ'R' \\ &= m_7 + m_6 + m_2 + m_4 \end{aligned}$$

Ans. (a)

- 50.
- P
- is a 16-bit signed integer. The 2's complement representation of
- P
- is
- $(F87B)_{16}$
- . The 2's complement representation of
- $8 \times P$
- is

- (a) $(C3D8)_{16}$ (b) $(187B)_{16}$
 (c) $(F878)_{16}$ (d) $(987B)_{16}$

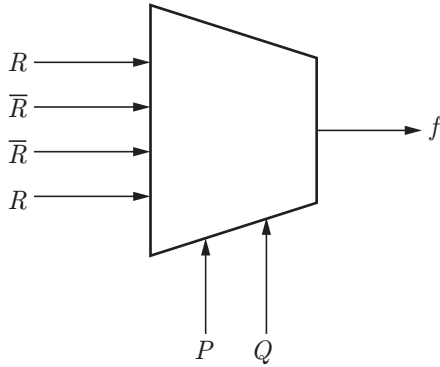
(GATE 2010: 1 Mark)

Solution:

$$\begin{aligned} (F87B)_{16} &= (1925)_2 \\ 1925 \times 8 &= (15400)_2 = 2\text{'s complement of } (15400)_2 \\ &= (C3D8)_{16} \end{aligned}$$

Ans. (a)

51. The Boolean expression for the output
- f
- of the multiplexer shown below is



- (a) $\overline{P \oplus Q \oplus R}$ (b) $P \oplus Q \oplus R$
 (c) $\overline{P \oplus Q \oplus R}$ (d) $P + Q + R$

(GATE 2010: 1 Mark)

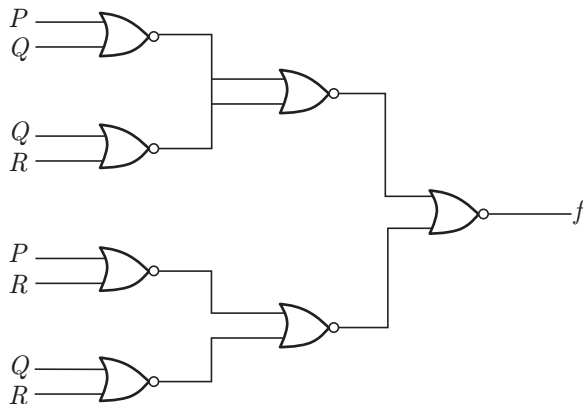
Solution:

P	Q	Input	Output
0	0	R	$P'Q'R$
0	1	R'	$P'QR'$
1	0	R'	$PQ'R'$
1	1	R	PQR

$$P'Q'R + P'QR' + PQ'R' + PQR = P \oplus Q \oplus R$$

Ans. (b)

52. What is the Boolean expression for the output f of the combinational logic circuit of NOR gates given below?



- (a) $\overline{Q + R}$ (b) $\overline{P + Q}$
 (c) $\overline{P + R}$ (d) $\overline{P + Q + R}$

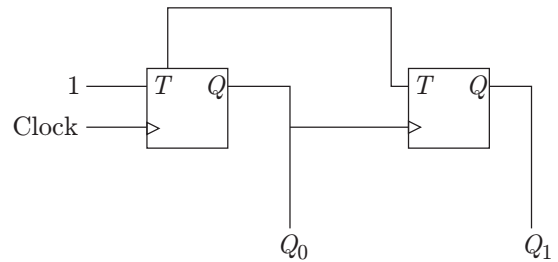
(GATE 2010: 2 Marks)

Solution:

$$\begin{aligned} & \overline{P + Q + Q + R + P + R + Q + R} \\ &= \overline{(P + Q + Q + R) \cdot (P + R + Q + R)} \\ &= \overline{(PQ + QR)(PR + QR)} \\ &= \overline{PQPR + PQQR + QRPR + QRQR} \\ &= \overline{PQR + PQR + PQR + QR} \\ &= \overline{PQR + QR} \\ &= \overline{QR(P + 1)} = \overline{QR} = \overline{Q + R} \end{aligned}$$

Ans. (a)

53. In the sequential circuit shown below, if the initial value of the output $Q_1 Q_0$ is 00, what are the next four values of $Q_1 Q_0$?



- (a) 11,10,01,00 (b) 10,11,01,00
 (c) 10,00,01,11 (d) 11,10,00,01

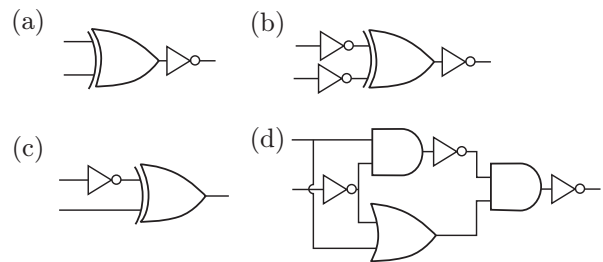
(GATE 2010: 2 Marks)

Solution:

Clock	Q_1	Q_0
1	0	0
2	1	1
3	1	0
4	0	1
5	0	0

Ans. (a)

54. Which one of the following circuits is NOT equivalent to a 2-input XNOR (exclusive NOR) gate?



(GATE 2011: 1 Mark)

Solution: Option (d) does not show XNOR gate functionality.

Ans. (d)

55. The simplified SOP (Sum of Product) form of the Boolean expression

$$(P + \bar{Q} + \bar{R}) \cdot (P + \bar{Q} + \bar{R}) \cdot (P + Q + \bar{R}) \text{ is}$$

- (a) $(\bar{P} \cdot Q + \bar{R})$ (b) $(P + \bar{Q} \cdot \bar{R})$
 (c) $(\bar{P} \cdot Q + R)$ (d) $(P \cdot Q + R)$

(GATE 2011: 1 Mark)

Solution:

$P \backslash QR$	00	01	11	10
0	0	1	1	1
1	0	0	0	0

which implies $(P + \bar{R})(P + \bar{Q})$

Ans. (b)

56. The minimum number of D flip-flops needed to design a Mod-258 counter is

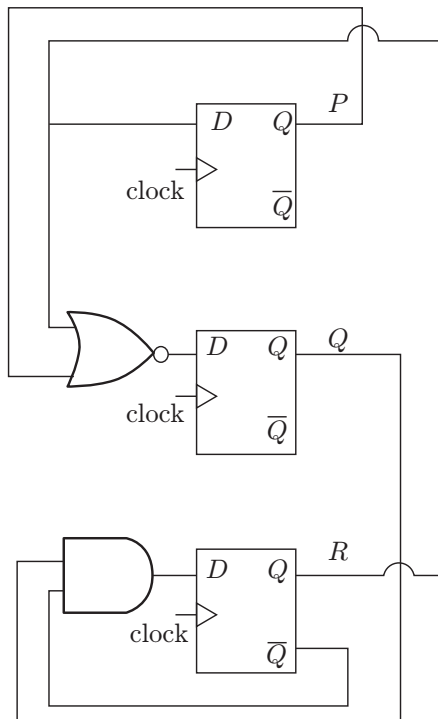
- (a) 9 (b) 8
 (c) 512 (d) 258

(GATE 2011: 1 Mark)

Solution: $2^n \geq 258 = 9 \text{ K flip-flops}$

Ans. (a)

Common Data Questions 57 and 58: Consider the following circuit involving three D -type flip-flops used in a certain type of counter configuration.



57. If at some instance prior to the occurrence of the clock edge, P , Q and R have a value 0, 1 and 0, respectively, what shall be the value of PQR after the clock edge?

- (a) 000 (b) 001
 (c) 010 (d) 011

(GATE 2011: 2 Marks)

Solution:

Clock	$D_1 = R$	$D_2 = (P + R)'$	$D_3 = QR'$	P	Q	R
1	0	1	0	0	1	0
2	0	1	1	0	1	1
3	1	0	0	1	0	0
4	0	0	0	0	0	0

After 010, value of PQR is 011.

Ans. (d)

58. If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by PQR generated by the counter?

- (a) 3 (b) 4
 (c) 5 (d) 6

(GATE 2011: 2 Marks)

Solution: Clock cycles can be seen from the table given in Question 57. The number of distinct outputs is 4.

Ans. (b)

59. The truth table

X	Y	$f(X, Y)$
0	0	0
0	1	0
1	0	1
1	1	1

represents the Boolean function

- (a) X (b) $X + Y$
 (c) $X \oplus Y$ (d) Y

(GATE 2012: 1 Mark)

Solution: From given truth table, equation formed is

$$XY' + XY \Rightarrow X(Y' + Y) \Rightarrow X$$

Ans. (a)

60. The decimal value 0.5 in IEEE single-precision floating-point representation has

- (a) Fraction bits of 000...000 and exponent value of 0
 (b) Fraction bits of 000...000 and exponent value of -1

- (c) Fraction bits of 100...000 and exponent value of 0
 (d) No exact representation

(GATE 2012: 1 Mark)

Solution: Binary representation of 0.5 is 0.1000, which can be written as 1.000×2^{-1} . Here, exponent = -1 and fraction bits are 000...000.
 Ans. (b)

61. What is the minimal form of the Karnaugh map shown below? Assume that **X** denotes a don't care term.

cd \ ab	ab			
	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

- (a) $\bar{b}\bar{d}$ (b) $\bar{b}\bar{d} + \bar{b}\bar{c}$
 (c) $\bar{b}\bar{d} + a\bar{b}\bar{c}d$ (d) $\bar{b}\bar{d} + \bar{b}\bar{c} + \bar{c}\bar{d}$

(GATE 2012: 2 Marks)

Solution:

	X	X	

From above grouping, the equation obtained is:

$$\bar{c}\bar{b} + \bar{d}\bar{b}$$

Ans. (b)

62. The smallest integer that can be represented by an 8-bit number in 2's complement form is

- (a) -256 (b) -128
 (c) -127 (d) 0

(GATE 2013: 1 Mark)

Solution: Smallest integer for 2's complement is -2^{n-1} .
 So here $n = 8$, $-2^{8-1} = -128$

Ans. (b)

63. In the following truth table, $V = 1$ if and only if the input is valid.

Inputs				Outputs		
D ₀	D ₁	D ₂	D ₃	X ₀	X ₁	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

What function does the truth table represent?

- (a) Priority encoder (b) Decoder
 (c) Multiplexer (d) Demultiplexer

(GATE 2013: 1 Mark)

Solution: From the given truth table it is clear that it is 4:2 priority encoder.

Ans. (a)

64. Which one of the following expressions does NOT represent exclusive NOR of x and y ?

- (a) $xy + x'y'$ (b) $x \oplus y'$
 (c) $x' \oplus y$ (d) $x' \oplus y'$

(GATE 2013: 1 Mark)

Solution: $x' \oplus y' = (x'y + y'x) = x \oplus y$, this represents XOR. So, option (d) is not exclusive NOR.

Ans. (d)

65. A RAM chip has a capacity of 1024 words of 8 bits each ($1K \times 8$). The number of 2×4 decoders with enable line needed to construct a $16K \times 16$ RAM from $1K \times 8$ RAM is

- (a) 4 (b) 5 (c) 6 (d) 7

(GATE 2013: 2 Marks)

Solution: Number of chips required = $\frac{16 \times 16}{1 \times 8} = 16 \times 2$

So, it requires 16 vertical and 2 horizontal chips. For this five 2×4 decoders are required.

Ans. (b)

PRACTICE EXERCISES

Set 1

1. Find the value of
- x
- in
- $(257)_{\text{base } 10} = (41)_{\text{base } x}$

(a) 61 (b) 64
(c) 127 (d) 256

2. What is the result of
- $01A2 + 001A = ?$

(a) 01BC (b) 01CB
(c) 01CC (d) 01DC

3. The simplified function of Boolean function
- $F(W, X, Y, Z) = \Sigma(0, 1, 2, 5, 8, 9, 10)$
- is

(a) $(W' + X')(Y' + Z')(X' + Z)$.
(b) $(W + X)(Y' + Z')(X' + Z')$.
(c) $(W' + X')(Y + Z)(X + Z)$.
(d) $(W' + X)(Y + Z')(X' + Z')$.

4. What is the hexadecimal representation of
- $(1356)_{\text{base } 8}$
- ?

(a) 2FE (b) 2DE
(c) 2EE (d) EEE

5. How many
- 3×8
- decoders are needed to construct
- 4×16
- decoder?

(a) 2 (b) 4
(c) 3 (d) 24/64

6. If 73 (in base
- x
- number system) is equal to 54 (in base
- y
- number system), the possible values of
- x
- and
- y
- are

(a) 11, 8 (b) 8, 13
(c) 11, 11 (d) 8, 11

7. Total number of
- 8×1
- MUX required for implementation of
- 256×1
- MUX is

(a) 37. (b) 35.
(c) 36. (d) 38.

8. Number of
- 8×1
- MUX needed at second last level is (assume level having only
- $1(8 \times 1)$
- MUX to be last level)

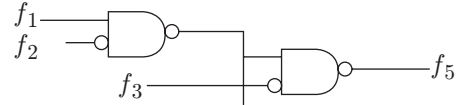
(a) 1 (b) 2
(c) 8 (d) 4

9. Match the columns

Column I	Column II
I. $xy + z(x \text{ XOR } y)$	A. $\Sigma m(1, 2, 4, 7)$
II. $x \text{ (XOR) } y \text{ (XOR) } z$	B. $\Sigma m(3, 5, 6, 7)$
III. $xy + yz + zx$	C. $\Sigma m(1, 2, 5, 6)$

(a) I-A, II-C, III-B (b) I-B, II-A, III-B
(c) I-A, II-B, III-A (d) I-A, II-C, III-B

10. Consider the following circuit:



Given: $f_1(x, y, z) = \Sigma m(1, 2, 6, 7)$, $f_3(x, y, z) = \Sigma m(0, 3)$, $f_5(x, y, z) = \Sigma m(0, 1, 3, 6)$

How many different minterms combinations are possible for f_2 ?

(a) 15 (b) 12
(c) 8 (d) 6

11. The two numbers represented in signed 2's complement form is
- $A = 11101101$
- and
- $B = 11100110$
- . If
- B
- is subtracted from
- A
- , the value obtained in signed 2's complement form is

(a) 10000111 (b) 00000111
(c) 10001001 (d) 11001001

12. Simplify the Boolean function

$$F(w, x, y, z) = \Sigma(1, 3, 10) + \Sigma d(0, 2, 8, 12)$$

(a) $F = w'x' + x'z$ (b) $F = wx' + x'z'$
(c) $F = w'x + x'z'$ (d) $F = w'x' + x'z'$

13. Which of the following is the complement of function
- $F = x(y'z' + yz)$

(a) $x + yz' + y'z'$ (b) $x' + yz' + y'z$
(c) $x' + y'z' + y'z$ (d) $x' + y'z' + y'z'$

14. Match the following:

IC Family	Characteristics
P. TTL	(i) High component density
Q. ECL	(ii) Low power consumption
R. MOS	(iii) Evolution of "diode transistor logic"
S. CMOS	(iv) High-speed digital circuits

P Q R S
(a) (i) (ii) (iii) (iv)
(b) (i) (iii) (iv) (ii)
(c) (iii) (iv) (i) (ii)
(d) (iv) (iii) (ii) (i)

15. What is the base value of binary number system?

(a) 0 (b) 2
(c) 1 (d) 01

16. What is the maximum value that we can represent using N bits?

- (a) $2n$ (b) 2^{n-1}
(c) $2^n - 2$ (d) $2^n - 1$

17. The number of bits required to represent 128 different numbers is

- (a) 5. (b) 7.
(c) 6. (d) 8.

18. Consider the following truth table, where A , B , C represents the inputs and F the output function

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

- (a) $B'(A + C')(A' + C')$ (b) $B(A + C')(A' + C)$
(c) $B'(A' + C')(A' + C)$ (d) $B(A + C)(A' + C')$

19. Which of the following expressions is equivalent to $A \oplus B \oplus C$?

- (a) $(A + B + C)(A' + B' + C')$
(b) $(A + B + C)(A' + B' + C)$
(c) $ABC + A'(B \oplus C) + B'(A \oplus C)$
(d) $(A' + B + C)(A' + B' + C')$

20. A half adder is also known as

- (a) NOR circuit. (b) XNOR circuit.
(c) NAND circuit. (d) XOR circuit.

21. Which of the following is minimum error code?

- (a) ANSI code (b) Grey code
(c) XS3 code (d) ASCII code

22. Register is a

- (a) set of capacitors and inductors used to provide input instructions in a digital computer
(b) set memory cells used to store the I/O data.
(c) temporary storage unit within the CPU.
(d) permanent storage unit within the CPU.

23. A T FF can be realized by SR FF by?

- (a) Connecting S to Q' .
(b) Connecting R is connected to Q' .
(c) Connecting S and R to Q .
(d) S and R are connected to Q and Q' , respectively.

24. Which FF has its output clearly defined on all the combinations inputs?

- (a) SR FF (b) D FF
(c) JK FF (d) T FF

25. The main difference between JK and RS flip-flop is that

- (a) JK flip-flop do not need a clock pulse whereas RS flip-flop requires the same.
(b) The output Q is connected to J and K is grounded whereas it is opposite in R and S .
(c) JK flip-flop accepts both inputs as 1.
(d) JK flip-flop is acronym of Johnson–Keith's multivibrator and RS of reset–set multivibrator.

26. AND gate provides a LOW output _____.

- (a) when any one input is LOW
(b) always
(c) when both the inputs are HIGH
(d) when both the inputs are LOW

27. How many pins does the 4049 IC have?

- (a) 22 (b) 16
(c) 20 (d) 40

28. Which of the following expressions is in the sum-of-products (SOP) form?

- (a) $A'B + C'D'$ (b) $AB'(CD)$
(c) $(A + B')(C + D)$ (d) $(A)B'(C + D')$

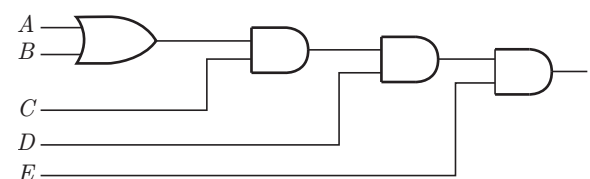
29. The significance of MSD in a decimal number system is

- (a) the middle digit of a stream of numbers.
(b) the digit to the right of the decimal point.
(c) the last digit on the right.
(d) the digit with the most weight.

30. The Generic Array Logic (GAL) device is _____.

- (a) complement of PAL.
(b) reprogrammable device.
(c) CMOS device.
(d) CMOS device which is also reprogrammable.

31. Derive the Boolean expression for the logic circuit shown below:



- (a) $C(A + B)DE$ (b) $C(A + B)D + E$
(c) $ABCDE$ (d) $(DE) + (ABC)$

32. An XNOR gate gives a HIGH output with the following input combination
- A = LOW, B = HIGH
 - A = HIGH, B = LOW
 - A = LOW, B = LOW
 - It is not possible to get a HIGH output in this gate
33. A shift register that has a parallel input, or a bidirectional serial load and having internal shift features, can be termed as
- Tristate device.
 - Parallel-in shift-out (PISO) register.
 - Universal shift register.
 - Bidirectional shift register.
34. Which of the following logic families has the least propagation delay?
- 54LYY
 - BiCMOS
 - ECL
 - 74SXX
35. Convert binary 11111110010 into hexadecimal.
- (EF2)₁₆
 - (FF2)₁₆
 - (FE2)₁₆
 - (FD2)₁₆
36. Convert the binary number (1001.0010)₂ to decimal.
- 90.125
 - 9.125
 - 91.25
 - 12.125
37. The alphanumeric code widely used for input and output to/from a computer?
- Biquinary Code
 - ASCII
 - ANSI Code
 - EBCDIC
38. (8B3F)₁₆ = (?)₂
- 111101101111
 - 01110101111
 - 101100111101111
 - 1000101100111111
39. Solving $-11 + (-2)$ will yield which 2's complement answer?
- 10011001
 - 1000 1001
 - 1111 0011
 - 1110 1001
40. Dividing (1100010)₂ by (0101)₂ yields (X)₁₀. The value of X is _____.
41. The solution to (0110)₁₆ + (10010)₁₆ = (?)₁₆
- 10120₁₆
 - 11120₁₆
 - 10020₁₆
 - 00B0₁₆
42. Which of the following is correct for full adders?
- Full adders can be used for multiplication also.
 - Full adders are used to make half adders and quad adders.
 - Full adders have three inputs including a carry input.
 - In a parallel full adder, the first stage may be a half adder.

43. What combination of the inputs a JK flip-flop toggles?
- $J = 0, K = 0$
 - $J = 0, K = 1$
 - $J = 1, K = 0$
 - $J = 1, K = 1$
44. In a 6-bit Johnson counter sequence, the total number of states or bit patterns is _____.

Set 2

1. The simultaneous equations on the Boolean variables x, y, z and w , are

$$\begin{aligned}x + y + z &= 1 \\xy &= 0 \\xz + w &= 1 \\xy + \bar{z}\bar{w} &= 0\end{aligned}$$

have the following solution for x, y, z and w , respectively.

- 0100
- 1101
- 1011
- 1000

2. Given the following Karnaugh map, which one of the following represents the minimal sum-of-product of the map?

$\begin{smallmatrix} wx \\ yz \end{smallmatrix}$	00	01	11	10
00	0	X	0	X
01	X	1	X	1
11	0	X	1	0
10	0	1	X	0

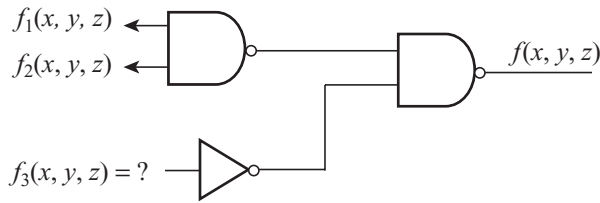
- $xy + y'z$
- $wx'y' + xy + xz$
- $w'x + y'z + xy$
- $xz + y$

3. Minimum SOP for $f(w, x, y, z)$ shown in Karnaugh-map below is

$\begin{smallmatrix} wx \\ yz \end{smallmatrix}$	00	01	11	10
00	0	1	1	0
01	X	0	0	1
11	X	0	0	1
10	0	1	1	X

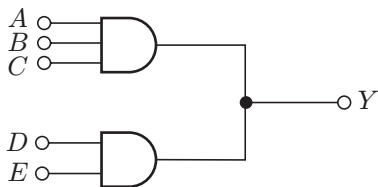
- $xz + y'z$
- $xz' + zx'$
- $x'y + zx'$
- None of the above

4. Consider the following logic circuit whose inputs are functions f_1, f_2, f_3 and output f .



Given that $f_1(x, y, z) = \sum(0, 1, 3, 5)$; $f_2(x, y, z) = \sum(6, 7)$ and $f_3(x, y, z) = \sum(1, 4, 5)$ then f is

- (a) $\sum(1, 4, 5)$ (b) $\sum(6, 7)$
 (c) $\sum(0, 1, 3, 5)$ (d) $\sum(4, 5, 7)$
5. $f(A, B) = A' + B$. Simplified expression for function $f(f(x + y, y), z)$ is
- (a) $x' + z$ (b) xyz
 (c) $xy' + z$ (d) None of the above
6. Two NAND gates having open collector outputs are tied together as shown in the following figure. The logic function Y , implemented by the circuit is



- (a) $Y = \bar{A}\bar{B}\bar{C} + \bar{D}\bar{E}$ (b) $Y = ABC + DE$
 (c) $Y = \bar{A}\bar{B}\bar{C} \cdot \bar{D}\bar{E}$ (d) $Y = ABC \cdot DE$
7. What is the equivalent Boolean expression in product-of-sums form for the Karnaugh map given in the following figure.

CD \ AB				
	00	01	11	10
00		1	1	
01	1			1
11	1			1
10		1	1	

- (a) $B\bar{D} + \bar{B}D$
 (b) $(B + \bar{C} + D)(\bar{B} + C + \bar{D})$
 (c) $(B + D)(\bar{B} + \bar{D})$
 (d) $(B + \bar{D})(\bar{B} + D)$

8. Which of the following sets of component(s) is/are sufficient to implement any arbitrary Boolean function?

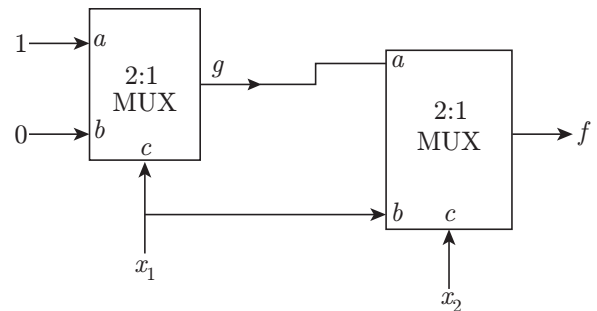
- (a) XOR gates, NOT gates
 (b) 2 to 1 Multiplexers
 (c) AND gates, XOR gates
 (d) Three-input gates that output $(A \cdot B) + C$ for the inputs A, B and C

9. Which function does NOT implement the Karnaugh map given in the following figure?

$wz \rightarrow$	00	01	11	10
$xy \downarrow$				
00	0	x	0	0
01	0	x	1	1
11	1	1	1	1
10	0	x	0	0

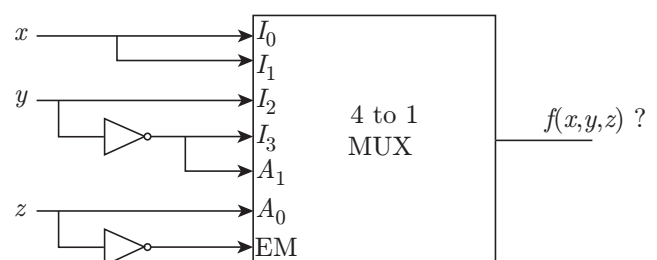
- (a) $(w + x)y$ (b) $xy + yw$
 (c) $(w + x)(w + y)(x + y)$ (d) None of the above

10. Consider the circuit shown below. The output of a 2:1 multiplexer is given by the function f ?



- (a) $x_1'x_2' + x_1x_2$ (b) $x_1'x_2 + x_1x_2'$
 (c) $x_1' + x_2$ (d) $x_1' + x_2'$

11. Consider the following multiplexer where I_0, I_1, I_2, I_3 are four data input lines selected by two address line combinations $A_1A_0 = 00, 01, 10, 11$, respectively, and f is the output of the multiplexer, EN is the Enable input.



The function $f(x, y, z)$ implemented by the above circuit is

- (a) xyz' (b) $xy + z$
(c) $x + y$ (d) None of the above

12. Which of the given number has its IEEE-754 32-bit floating-point representation as (0 10000000 110 0000 0000 0000 0000 0000)

- (a) 2.5 (b) 3.0
(c) 3.5 (d) 4.5

13. Consider the following state table in figure for a sequential machine. The number of states in the minimized machine will be

	0	1
A	D, 0	B, 1
B	A, 0	C, 1
C	A, 0	B, 1
D	A, 1	C, 1

- (a) 4 (b) 3
(c) 2 (d) 1

14. A ROM is used to store the table for multiplication of two 8-bit unsigned integers. The size of ROM required is

- (a) $256K \times 16$ (b) $64K \times 8$
(c) $4K \times 16$ (d) $64K \times 16$

15. Booth's algorithm for integer multiplication gives worst performance when the multiplier pattern is

- (a) 101010.....1010 (b) 100000.....001
(c) 111111.....1111 (d) 011111.....1110

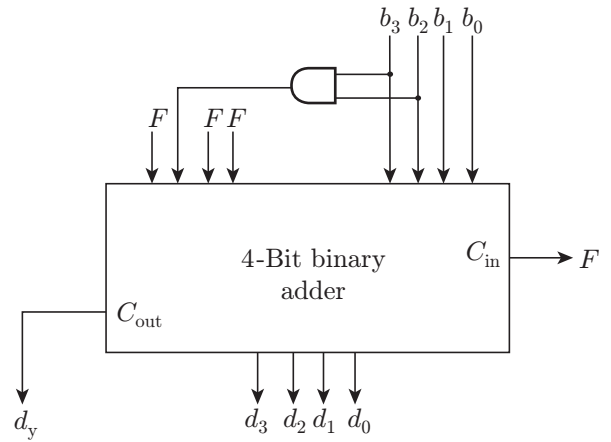
16. Consider the following floating point number representation.

31	24	23	0
Exponent		Mantissa	

The exponent is in 2's complement representation and mantissa is in the sign magnitude representation. The range of the magnitude of the normalized numbers in this representation is

- (a) 0 to 1 (b) 0.5 to 1
(c) 2^{-23} to 0.5 (d) 0.5 to $(1 - 2^{-23})$

17. Consider the circuit in the following figure which has a 4-bit binary number $b_3b_2b_1b_0$ as input and a 5-bit binary number $d_4d_3d_2d_1d_0$ as output. The circuit implements



- (a) Binary to Hex conversion
(b) Binary to BCD conversion
(c) Binary to grey code conversion
(d) Binary to radix -12 conversion

18. Given $\sqrt{(224)_r} = (13)_{r'}$

The value of the radix r is

- (a) 10 (b) 8 (c) 5 (d) 6

19. Zero has two representations in:

- (a) sign magnitude (b) 1's complement
(c) 2's complement (d) both (a) and (b)

20. The number 43 in 2's complement representation is

- (a) 01010101 (b) 11010101
(c) 00101011 (d) 10101011

21. Consider the values $A = 2.0 \times 10^{30}$, $B = -2.0 \times 10^{30}$, $C = 1.0$ and the sequence

$$X := A + B$$

$$Y := A + C$$

$$X := X + C$$

$$Y := Y + B$$

- (a) $X = 1.0$, $Y = 1.0$ (b) $X = 0.0$, $Y = 1.0$
(c) $X = 1.0$, $Y = 0.0$ (d) $X = 0.0$, $Y = 0.0$

22. The 2's complement representation of $(-539)_{10}$ in hexadecimal is

- (a) ABE (b) DBC
(c) DE5 (d) 9E7

23. The decimal value 0.25

- (a) is equivalent to binary 0.1
(b) is equivalent to binary 0.01
(c) is equivalent to binary 0.00111
(d) cannot be represented precisely in binary

24. Sign extension is the step in

- (a) floating point multiplication.
(b) signed 16-bit integer addition.

- (c) arithmetic left shift.
 (d) converting a signed integer from one size to another.
25. The state of a 12-bit register is 100010010111. It is represented in BCD, what is its content?
 (a) 2199 (b) 3423
 (c) 564 (d) 897
26. Simplification of given Boolean expressions = $\{(a + b' + c')(a' + c')''\}$ to a minimum number of literals is
 (a) $(c' + a'b')$ (b) $(a' + b'c')$
 (c) $(b' + a'c')$ (d) $(a' + b' + c')$
27. Simplification of the following Boolean function $\{F(A, B, C, D) = \sum(3, 7, 11, 13, 14, 15)\}$ using K-maps:
 (a) $CD + ABC + BCD$ (b) $CD + ABC + ABD$
 (c) $CD + ACD + ABD$ (d) $CD + ABC + ACD$
28. A flip-flop has a 4-ns delay from the time the clock edge occurs to the time the output is complemented in a 10-binary ripple counter. The ripple counter operates reliably at which maximum frequency?
 (a) 25 MHz (b) 30 MHz
 (c) 32 MHz (d) 40 MHz
29. How many parity check bits must be included with the data word to achieve single-error correction when data word contains 16 bits?
 (a) 4 Bits (b) 5 Bits
 (c) 6 Bits (d) 7 Bits
30. How many $32K \times 8$ RAM chips are needed to provide a memory capacity of 256K bytes?
 (a) 4 (b) 5
 (c) 6 (d) 8

ANSWERS TO PRACTICE EXERCISES

Set 1

- | | | | | | |
|--------|---------|---------|---------|---------|----------|
| 1. (b) | 9. (b) | 17. (b) | 25. (c) | 33. (c) | 41. (a) |
| 2. (a) | 10. (a) | 18. (d) | 26. (a) | 34. (c) | 42. (d) |
| 3. (a) | 11. (b) | 19. (c) | 27. (b) | 35. (b) | 43. (d) |
| 4. (c) | 12. (d) | 20. (c) | 28. (a) | 36. (b) | 44. (12) |
| 5. (a) | 13. (b) | 21. (b) | 29. (d) | 37. (b) | |
| 6. (d) | 14. (c) | 22. (c) | 30. (b) | 38. (d) | |
| 7. (a) | 15. (b) | 23. (d) | 31. (a) | 39. (c) | |
| 8. (d) | 16. (d) | 24. (c) | 32. (c) | 40. (3) | |

Set 2

1. (c) Considering each option
 (a) Given $x = 0, y = 1, z = 0$ and $w = 0$
 i. $x + y + z = 0 + 1 + 0 = 1$
 ii. $xy = 0 \cdot 1 = 0$
 iii. $xz + w = 0 \cdot 0 + 0 = 0$
 but $xz + w$ should be 1, so option (a) is incorrect.
 (b) Given $x = 1, y = 1, z = 0$ and $w = 1$
 i. $x + y + z = 1 + 1 + 0 = 1$
 ii. $xy = 1 \cdot 1 = 1$

However, given $xy = 0$
 Therefore, option (b) is not the solution.

- (c) Given $x = 1, y = 0, z = 1$ and $w = 1$
 i. $x + y + z = 0 + 0 + 1 = 1$
 ii. $xy = 1 \cdot 0 = 0$
 iii. $xz + w = 1 \cdot 1 + 1 = 1$
 iv. $xy + z'w' = 1 \cdot 0 + 0 \cdot 0 = 0$

It satisfies all the given conditions. So, (c) is the correct option.

2. (a)

$wx \backslash yz$	00	01	11	10
00	0	X	0	X
01	X	1	X	1
11	0	X	1	0
10	0	1	X	0

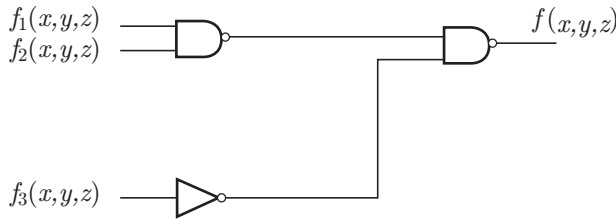
SOP is $xy + y'z$.

3. (c)

$wx \backslash yz$	00	01	11	10
00	0	1	1	1
01	x	0	0	0
11	x	0	0	0
10	0	1	1	x

Sum-of-product for given function is $x'y + zx'$.

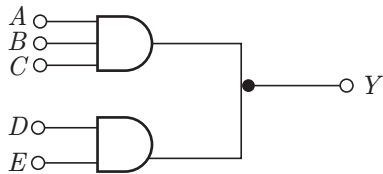
4. (a)



$$\begin{aligned}
 f_1 f_2 &= f_1(x, y, z) \text{ NAND } f_2(x, y, z) \\
 &= \sum (0, 1, 2, 3, 4, 6, 7) \\
 f_3 &= \sum (1, 4, 5) \\
 f_3' &= \sum (0, 1, 2, 3, 6, 7) \\
 f &= f_1 f_2 \text{ NAND } f_3' = (f_1 f_2 f_3')' \\
 &= (\sum (0, 1, 2, 3, 6, 7))' = \sum (1, 4, 5)
 \end{aligned}$$

$$\begin{aligned}
 5. (c) \quad f(f(x+y, y), z) &= f((x+y)' + y, z) = f(x'y' + y, z) \\
 &= f(x' + y, z) = (x' + y)' + z = xy' + z
 \end{aligned}$$

6. (c)



$$\text{Therefore, } Y = \bar{A}\bar{B}\bar{C} \cdot \bar{D}\bar{E}$$

7. (c) The Karnaugh map for POS can be given by

$CD \backslash AB$	00	01	11	10
00	0	0	1	1
01	X	X	1	x
11	1	1	1	0
10	0	1	1	0

POS form of f is given as
 $F(C, D, A, B) = (B + D)(\bar{B} + \bar{D})$

8. (c) Functionally complete operation set is the set of logic functions from which all other functions can be realized.

[XOR, AND] is a functionally complete set.

9. (c) For SOP form

$wz \backslash xy$	00	01	11	10
00	0	X	0	1
01	0	X	1	1
11	1	1	1	1
10	0	X	0	0

f in SOP can be given as

$$f(x, y, w, z) = xy + yw$$

or

$$y(x + w)$$

Therefore, options (a) and (b) are true.

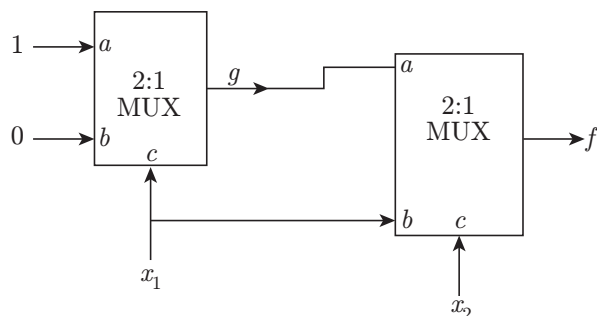
For POS form

$wz \backslash xy$	00	01	11	10
00	0	X	0	0
01	0	X	1	1
11	1	1	1	1
10	0	X	0	0

$$f(x, y, w, z) = y(x + w)$$

Option (c) is incorrect.

10. (a)



The output g is $g = 1 \cdot x_1' + 0 \cdot x_1 = x_1'$

The output f is

$$f = gx_2' + x_1x_2$$

$$f = x_1'x_2' + x_1x_2$$

11. (a)

$$= \text{EN}(\bar{A}_1\bar{A}_0I_0 + \bar{A}_1A_0I_1 + A_1\bar{A}_0I_2 + A_1A_0I_3)$$

$$= z'(xyz' + yzx + 0 + y'zy')$$

$$= xyz' + xyz' + 0 \cdot z' + z' \cdot zy'$$

$$= xyz' + 0 + 0 + 0 = xyz'$$

12. (c) Sign bit $S = 0 \Rightarrow$ positive number

$$E = 1000\ 0000\text{B} = 128\text{D (in normalized form)}$$

$$\text{Fraction is } 1.11\text{B (with an implicit leading 1)}$$

$$= 1 + 1 \times 2^{-1} + 1 \times 2^{-2} = 1.75\text{D}$$

$$\text{The number is } +1.75 \times 2^{(128-127)} = +3.5\text{D}$$

13. (b)

	x	0	1
A		$D, 0$	$B, 1$
B		$A, 0$	$C, 1$
C		$A, 0$	$B, 1$
D		$A, 1$	$C, 1$

These two states work same

Combining two states B and C into single state " BC ".

	x	0	1
A		$D, 0$	$BC, 1$
BC		$A, 0$	$BC, 1$
D		$A, 0$	$BC, 1$

Therefore, the number of states in the minimized machine will be 3.

14. (d) To store the table for multiplication of two 8-bit unsigned integers, size of ROM required is

$$2^{2N} \times 2N = 2^{16} \times 16 = 64\text{K} \times 16$$

15. (a)

Booth recording of a multiplier are

$$\begin{array}{cccccccccccccccc} \text{(a)} & 1 & 0 & 1 & 0 & 1 & 0 & \dots & 1 & 0 & 1 & 0 & & & & & \\ & \downarrow & & & & & & & & & & & & & & & \\ & -1 & +1 & -1 & +1 & -1 & +1 & \dots & -1 & +1 & -1 & 0 & & & & & \\ \text{(b)} & 1 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 & 0 & 1 & & & & & \\ & \downarrow & & & & & & & & & & & & & & & \\ & -1 & +0 & 0 & 0 & 0 & 0 & \dots & -0 & 0 & +1 & -1 & & & & & \\ \text{(c)} & 1 & 1 & 1 & 1 & 1 & 1 & \dots & 1 & 1 & 1 & 1 & & & & & \\ & \downarrow & & & & & & & & & & & & & & & \\ & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 & 0 & -1 & & & & & \\ \text{(d)} & 0 & 1 & 1 & 1 & 1 & 1 & \dots & 1 & 1 & 1 & 0 & & & & & \\ & \downarrow & & & & & & & & & & & & & & & \\ & 1 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 & -1 & 0 & & & & & \end{array}$$

Therefore, option (a) gives the worst performance because it contains maximum number of 1's.

16. (d)

31	24	23	0
Exponent		Mantissa	

The range of the magnitude of the normalized number: 0.5 to $(1 - 2^{-23})$

17. (d) Whenever both b_3 and b_2 is 1, then 01 00 is added to the number.

$$b_3b_2b_1b_0 = (1100)_B = (12)_{10}$$

So, in all the numbers greater than equal to 12, 0100 is added. Therefore, the circuit is binary to radix -12 converter.

18. (c)

(a) When the radix r is 10

$$\text{Then } \sqrt{(224)_{10}} \approx 14.967 \neq (13)_{10}$$

(b) When the radix is 8

$$\text{Then } \sqrt{(224)_8} \approx \sqrt{(148)_{10}} = (12.165)_{10} \text{ and}$$

$$(13)_8 = (11)_{10}. \text{ Hence, } \sqrt{(224)_8} \neq (13)_8$$

Option (b) is not correct.

(c) When the radix is 5

$$\sqrt{(224)_5} \approx \sqrt{(64)_{10}} = (8)_{10} \text{ and } (13)_5 = (8)_{10}.$$

$$\text{Therefore, } \sqrt{(224)_5} = (13)_5$$

Option (c) is correct.

(d) When the radix is 6

$$\sqrt{(224)_6} = \sqrt{(88)_{10}} \approx (9.38)_{10} \text{ and } (13)_6 = (9)_{10}.$$

Therefore, $\sqrt{(224)_6} \approx (13)_6$

Option (d) is not correct.

19. (d) Zero has two representations in

Sign magnitude as (i) 0 0 0 0, (ii) 1 0 0 0

1's Complement as (i) 0 0 0 0, (ii) 1 1 1 1

20. (c) Positive 2's complement numbers are represented as same simple binary. So +43 in binary is 0 0 1 0 1 0 1 1.

21. (a) Given that $A = 2.0 \times 10^{30}$, $B = -2.0 \times 10^{30}$, $C = 1.0$

$$(i) X = A + B = 2.0 \times 10^{30} + (-2.0 \times 10^{30}) = 0.0$$

$$(ii) Y = A + C = 2.0 \times 10^{30} + (1.0)$$

$$(iii) Y = X + C = 0.0 + 1.0 = 1.0$$

$$(iv) Y = Y + B = 2.0 \times 10^{30} + 1.0 + (-2.0 \times 10^{30}) = 1.0$$

Values of X and Y are 1.0 and 1.0, respectively.

22. (c) Compute a binary representation of the magnitude of the number and then take 2's complement of binary representation.

$$+539 = 0010\ 0001\ 1011$$

$$-539 = 1101\ 1110\ 0101$$

$$= (\text{DE5})_{16}$$

23. (b)

$$\begin{array}{rcl} 0.25 \times 2 = 0.50 & 0 & \uparrow \\ 0.50 \times 2 = 1.00 & 1 & \end{array}$$

Therefore, the decimal value 0.25 is equivalent to binary 0.01.

24. (d) Sign extension is the operation for increasing the number of bits of a binary number while preserving the number's sign (positive/negative) and value.

25. (d)

$$\begin{aligned} 26. (a) & (a + b' + c')(a' + c') \\ & \Rightarrow aa' + ac' + a'b' + b'c' + a'c' + c'c' \\ & \Rightarrow c' + b'(a' + c') \\ & \Rightarrow c' + a'b' \end{aligned}$$

27. (b)

		CD			
AB		00	01	11	10
		m ₀	m ₁	m ₈	m ₂
00				1	
01		m ₄	m ₅	m ₇	m ₆
11		m ₁₂	m ₁₃	m ₁₅	m ₁₄
10		m ₅	m ₉	m ₁₁	m ₁₀

ABD CD

← ABC

$$F = CD + ABC + ABD$$

28. (a) In worst case, all 10 flip-flops are complemented. The maximum delay is $10 \times 3 \text{ ns} = 30 \text{ ns}$. So, the maximum frequency is $10^9/40 = 25 \text{ MHz}$.

29. (b) Solving for n in terms of k , we obtain $2^k - 1 - k \geq n$

For data word length (n) = 16 bit we require 5 bit.

30. (d) $256/32 = 8$ chips